MSF

UNIT-1 :- Basics of Number theory.

UNIT-2 :- Random Variables and probability Distributions.

UNIT-3 :- continuous Distributions and Sampling.

UNIT-4 :- Test of Hypothesis (large and small Samples).

UNIT-5 :- Applied statistics.

2.5 Easy chap.

UNIT-2

RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

~ Theory of probability is one of the most Intresting Branches of modern mathematics.

~ Probability is used in many areas such as Insurance, statitics, Biology and physical science, Engineering, Business and Computer Science it helps as make better decisions when the outcome is uncertain.

~ The phrase "probable" is often used in daily conversation which means "likely".

~ The probability of happening of an event means that the amount of chance (likelyhood) of the happenig of that event.

~ If an Experiment is repeated under similar (and) Homogenous Conditions, we can see two types of situations.

i) The net Result outcome is unquie (or) Certain
i.e Determenistic (or) predictable

Ex: $V = \frac{4}{3}\pi r^3$

ii) The net result outcome is not unquie.
i.e unpredictable (or) probabilistic
Ex: rain comes or not

Random Experiment

An Experiment can be repeated any no: of times under the same Identical conditions and what are the total outcomes we should know advance.

Ex:- Tossing a coin,
Throughing a die,
picking a coin

Trial :-

Each performance in a random Experiment is called Trial.

Ex:- Tossing a coin

Sample Space :-

(Total no. of sample outcomes)

A set of all possible outcomes of an Experiment is called a Sample Space.

Ex:- In an Experiment of Tossing of a coin, possible outcomes are Head and Tail.

$\therefore$ Sample space 'S' = {H, T}

* In throwing a die S = {1, 2, 3, 4, 5, 6}

Event :-

Any Subset of a Sample space is called an Event.

→ It is denoted by 'E'.

* In an Experiment of tossing of a coin getting Head is an Event (or) getting tail is an Event.

① $S = \{H, T\}$

$E_1 = \{H\}$, $E_2 = \{T\}$

② $S = \{1, 2, 3, 4, 5, 6\}$

$E_1 = \{1\}$, $E_2 = \{4\}$, $E_3 = \{3 \text{ or } 4\}$

Equally likely Events :-

~ Outcomes of a trial of said to be Equally likely, if one cannot be Expected in performance to the others.

Ex:- ① Tossing a coin, Two possible cases (or) Equally likely Event.

② when card is drawn from a pack, Any card may be obtain in this trial all the 52 cards or Equally likely Events.

Mutually Exclusive Events :-

Two Events A and B are mutually Exclusive if $A \cap B = \phi$

Ex:- Tossing two coins,

A = {Two heads turning up} = {H, H}

B = {one Head turning up} = {HT, TH}

C = {No Head Turning up} = {TT}

A, B, C are mutually Exclusive Events.

NOTE:- ① If A, B, C are mutually Exclusive then $A \cap B = \phi$, $B \cap C = \phi$, $C \cap A = \phi$

② If $A_1, A_2, A_3 \ldots A_n$ are mutually Exclusive then $A_1 \cap A_2 \cap A_3 \cap \ldots \cap A_n = \phi$ and

$$\boxed{A_i \cap A_j = \phi} \quad i \neq j$$

Exhaustive Events:-

Total no. of all possible Outcomes of a random Experiment is known as Exhaustive Events.

Ex:- ① Tossing 2 coins, there are 4 exhaustive Events.

② Throwing 2 coins, the Exhaustive no. of Events. is $6^2 = 36$.

Favorable Events:-

~ The no. of cases favorable to an Event in a trial is the no. of Outcomes which Entail happening of the Event.

Ex:- In a throw of 2 dice, the no. of favorable Events to get in same numbers on 2 dice is '6'.

i,e $\{(1,1)\ (2,2)\ (3,3)\ (4,4)\ (5,5)\ (6,6)\}$

Probability:- If an Exp. is performed, n is the no. of Exhausted cases and 'm' is the favorable cases of an event 'E' then the probability of an Event 'E' is defined as

$$P(E) = \frac{\text{no. of favorable cases to an event } E}{\text{no. of Exhaustive cases}}$$

$$P(E) = \frac{n(E)}{n(S)}$$

Random Vairables.

Concept of Random Vairables.

A Random Vairables is a vairable whose values depend on the Outcome of the Random Exp.

Data —
- Measurable thing Ex:- getting no.
- Non-Measurable thing Ex:- pain.

pain, it is Quantitative data.

Qualitative data:-

It is cateogries into two types.

i) Discrete data / vairable.

Ex:- 1, 2, 3, 4, 5, 6

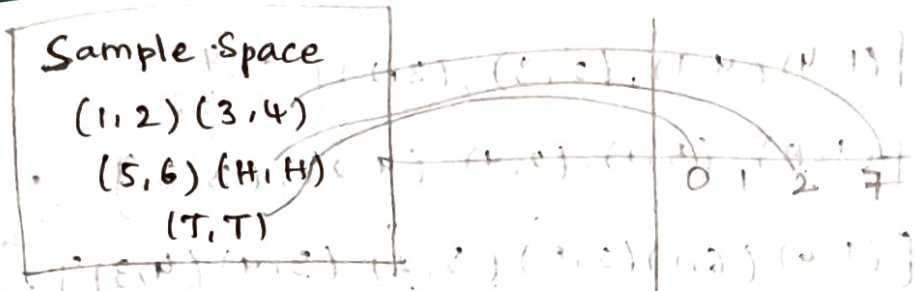
ii) continious vairable (a, b)

Ex:- Height, weight

= 17.3 cm ; 16.5 kg.

def:- A random vairable a function that design a real no. to Every Outcome to the sample Space of a Random Experiment.

— A R.V are usually denoted by capital letters such as X, Y, Z.



Ex+1) Suppose two coins are tossed simanteously then sample space (S) = {HH, TH, HT, TT}

let X = no. of heads then the (i) X=0 then the outcome is {TT} and $P(\{TT\}) = P(X=0) = \frac{1}{4}$

ii) X=1 then the outcome is {HT, TH} and

$P(\{HT\}, \{TH\}) = P(X=1) = \frac{3}{4} = \frac{1}{2}$

(a,b) = (a+b)

iii) if X=2, the outcome is {HH} and $P(\{HH\}) = P(X=2) = \frac{1}{4}$

⇒ The probability distribution of Random Vairable 'X' is given by the following table.

X = x	0	1	2
P(X=x)	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

2) Two dice are thrown. let X assign to each point (a,b) in S, the ~~maximum~~ sum of its numbers i.e X (a,b) = a+b. Find the probability distribution fun^n.

Sol+ w.K.T n(S) = 36

Given that X (a,b) = (a+b)

$P(X=2) = P((1,1)) = \frac{1}{36}$

$P(X=3) = P((1,2)(2,1)) = \frac{2}{36}$

$P(X=4) = P((1,3),(3,1)(2,2)) = \frac{3}{36}$

$P(X=5) = P[(1,4),(4,1),(2,3),(3,2)] = \frac{4}{36}$

$P(X=6) = P[(1,5),(5,1),(2,4),(4,2),(3,3)] = \frac{5}{36}$

$P(X=7) = P[(1,6)(6,1)(2,5)(5,2)(3,4)(4,3)] = \frac{6}{36}$

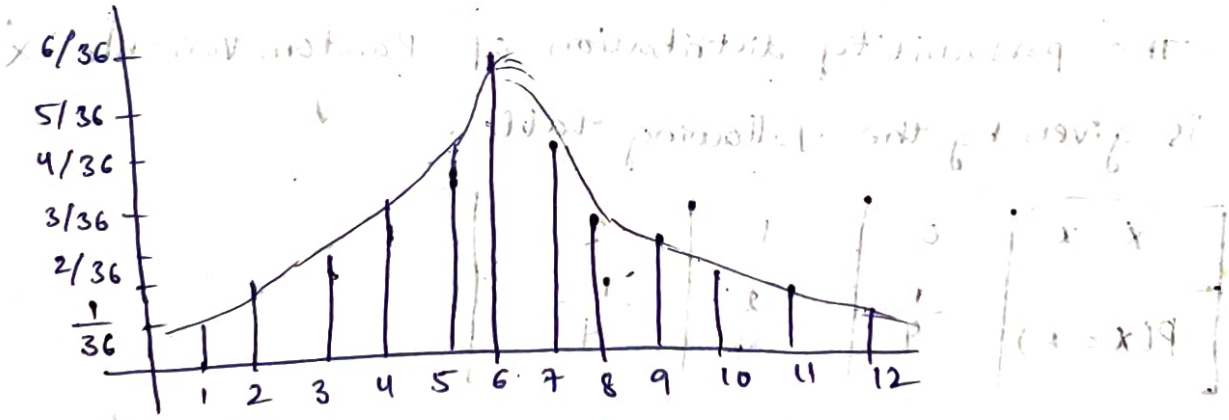
$P(X=8) = P[(2,6)(6,2)(3,5)(5,3)(4,4)] = \frac{5}{36}$

$P(X=9) = \frac{4}{36}$ | $P(X=11) = \frac{2}{36}$

$P(X=10) = \frac{3}{36}$ | $P(X=12) = \frac{1}{36}$.

The prob. distribution fun^n for r.v x is given by

$X=x$	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$.



Types of Random Vairables:

1) Discrete Random Vairables.

- A R.V is said to be discrete if it sample space (s) is contantable.

(or)

- A R.V that takes countable values:

Ex: 1) Human walking.

2) Horse riding

3) No. of students in a class.

4) No. of members in a family.

5) No. of defective bulbs in a sample of electric bulbs.

6) No. of printing mistakes in each page of a book.

7) No. of Telephone calls recieve by the telephone operator.

prob. distribution fun^n (or) Mass function :-

The set of ordered pairs $(x, p(x))$ are $(x, f(x))$ is called prob. distribution fun^n (or) mass fun^n. of the discrete R.V of X and is usually given the form

X	x_1	x_2	x_3 ----------	x_n
$P(X=x)$	$P(x=x_1)$	$P(x=x_2)$	$P(x=x_3)$	$P(x_n)$

$P(X)$ also statifies the following cond^{n}s :-

i) $P(x_i) \geq 0$

ii) $\sum_{i=1}^{n} P(x_i) = 1$

iii) $P(X=x) = f(x)$

problems :-

** 1) A R.V X has the following probability fun^n.

$X=x$	0	1	2	3	4	5	6
$P(X=x)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$

i) Find K

ii) Evaluate $P(X<4)$

iii) $P(X \geq 5)$

iv) $P(3 < x \leq 6)$

** v) what is the smallest value of x for which $P(x \leq x) > \frac{1}{2}$?

Soln W.K.T the total prob. is unit (1).

i) i.e $\sum_{i=1}^{n} p(x_i) = 1 \Rightarrow \sum_{i=1}^{6} p(x_i) = 1$

$$P(X=0) + P(X=1) + P(X=2) + \cdots + P(X=6) = 1$$

$$k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$49k = 1 \Rightarrow k = \frac{1}{49}$$

ii) $X < 4$

$$P(X<4) = P(X=0) + P(X=1) + P(X=2) + P(X=3) = 49$$

$$= k + 3k + 5k + 7k \Rightarrow 16k$$

$$\Rightarrow k = \frac{16}{49}$$

iii) $P(X \geq 5)$

$$\Rightarrow P(X=5) + P(X=6) \Rightarrow 11k + 13k = 49$$

$$= 24k$$

$$= k = \frac{24}{49}$$

iv) $P(3 < x \leq 6) = P(X=4) + P(X=5) + P(X=6) = 49$

$$= 9k + 11k + 13k = 33k$$

$$k = \frac{33}{49}$$

v) $P(X \leq x) > \frac{1}{2}$

The smallest value 'x' for which $P(X \leq x) > \frac{1}{2}$ we are 'x' determine but trial method. i.e Subsituting vairous values of 'x'.

for $x=6$, $P(X \leq 6) = P(0) + P(1) + P(2) + \cdots + P(6)$

$$= 49k \Rightarrow \frac{49}{49} = 1$$

$$= 1 > \frac{1}{2}$$

for $x=5$, $P(X \leq 5) = P(0) + P(1) + \cdots + P(5)$

$$= 36k$$

$$\Rightarrow k = 36/49 = 0.73 > \frac{1}{2}$$

for $x=4$, $P(X \leq 4) = p(0) + p(1) + \cdots + p(4)$

$= k + 3k + 5k + 7k + 9k$

$k = \frac{25}{49} = 0.51 > \frac{1}{2}$

for $x=3$, $P(X \leq 3) = p(0) + p(1) + p(2) + p(3)$

$= k + 3k + 5k + 7k$.

$= 16k \Rightarrow k = \frac{16}{49} = 0.32 \not> \frac{1}{2}$

∴ Smallest value of 'x' for which $P(X \leq x) > \frac{1}{2}$ is 4.

2) A Random vairable 'X' has a following prob. distribution. **

$X = x$	0	1	2	3	4	5	6	7
$P(X=x)$	0	k	$2k$	$2k$	$2k$	$3k$	k^2	$7k^2+k$

i) Find the value of k

ii) $P(X<6)$, $P(X \geq 6)$

iii) Evaluate $P(0 < x < 5)$

Soln- i) W.K.T the total prob. is 1

$$\sum_{i=1}^{6} p(x_i) = 1$$

$\Rightarrow P(X=0) + P(X=1) + \cdots + P(X=7) = 1$

$0 + k + 2k + 2k + 2k + 3k + k^2 + 7k^2 + k$

$8k^2 + 11k + 0 \Rightarrow k = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$= \frac{-11 \pm \sqrt{11^2 - 4(8)(0)}}{2(8)} \Rightarrow \frac{-11 \pm 3\sqrt{17}}{16}$

$k = 0.085$

ii) $P(x<6)$, $P(x \geq 6)$

$P(x<6) = p(x=0) + p(x=1) + p(x=2) + p(x=3) + p(x=4) + p(x=5)$

$= 0 + k + 2k + 2k + 2k + 3k$

$= 10k$

$= 10\{ + 0.085 = 0.85$

$P(x \geq 6)$:-

$P(x=0) + P(x=1) + \cdots + P(x=6) + P(x=7)$

$= 0 + k + 2k + 2k + 2k + 3k + k^2 + 7k^2 + k$

$= 11k + 8k^2$

$= 11(0.1) + 8(0.1)^2$

$= 0.14$

(iii) $P(0 < x < 5)$

$= P(x=1) + P(x=2) + P(x=3) + P(x=4)$

$= k + 2k + 2k + 2k$

$= 7k \Rightarrow 7(0.1) = 0.7$

Continous Random Vairables :-

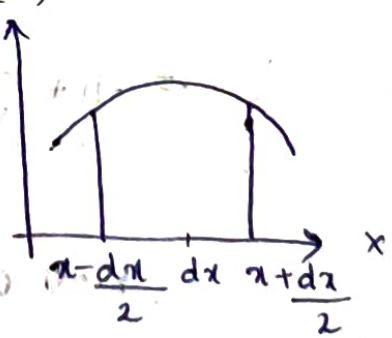
The R.V 'x' is said to be continious if the sample space (S) contain infinite numbers (or) A R.V that takes any value with an interval.

- take any values with in interval.

Ex:- Height, weight, Temparature and time, Speed.

probability density function :-

The curve $y = f(x)$ is known as probability density curve (or) simply probability curve.



(2, 3)

- The for continous R.V $[x, f(x)]$, is called probability density function defined over

set of real numbers. and also statifies the following conditions.

(i) $f(x_i) \geq 0$

(ii) $\sum_{i=1}^{n} p(x_i) = 1$

ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

iii) $p(a \leq x \leq b) = \int_a^b f(x)\,dx$

iv) $p(a < x \leq b) = p(a \leq x < b)$
$= p(a \leq x \leq b)$
$= \int_a^b f(x)\,dx$

*comp.

Q → A continous Random vairable 'X' has the prob.

i) density fun^n $f(x) = \begin{cases} kx^2 & ; 0 \leq x \leq 1 \\ 0 & ; \text{else where} \end{cases}$

find the value of k, with this value of k, find $p(x < \frac{1}{2})$ and $p(x \geq \frac{3}{4})$.

Sol:- W.K.T the total probability is unity (1)

$\therefore \int_{-\infty}^{\infty} f(x)\,dx = 1$

$\Rightarrow \int_{-\infty}^{0} f(x)\,dx + \int_0^1 f(x)\,dx + \int_1^{\infty} f(x)\,dx = 1$

$= \int_{-\infty}^{0} kx^2\,dx + \int_0^1 kx^2\,dx + \int_1^{\infty} kx^2\,dx = 1$

$= 0 + \int_0^1 kx^2\,dx + 0 = 1$

$\Rightarrow k\int_0^1 x^2\,dx = 1 \Rightarrow k\left(\frac{x^3}{3}\right)_0^1 = 1$

$k\left(\frac{1}{3}\right) = 1 \Rightarrow \boxed{k = 3.}$

$\therefore f(x) = \begin{cases} 3x^2 & ; 0 \leq x \leq 1 \\ 0 & ; \text{other wise} \end{cases}$

Now, $p\left(x < \frac{1}{2}\right) = \int_{-\infty}^{1/2} f(x)\,dx$

$$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1/2} f(x)\,dx$$

$$= 0 + \int_{0}^{1/2} f(x)\,dx \Rightarrow \int_{0}^{1/2} 3x^2\,dx$$

$$= 3\left(\frac{x^3}{3}\right)_0^{1/2} \Rightarrow \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Now $P\left(x \geq \frac{3}{4}\right) \Rightarrow P\left(\frac{3}{4} \leq \frac{x}{4}\right)$

$$= P\left(\frac{3}{4} \leq x\right) = \int_{3/4}^{\infty} f(x)\,dx$$

$$= \int_{3/4}^{1} f(x)\,dx = \int_{3/4}^{1} 3x^2\,dx$$

$$= \left(x^3\right)_{3/4}^{1} \Rightarrow \frac{37}{64}$$

2) Is the function $f(x)$ defined as follows, density function ? $f(x) = \begin{cases} 0 ; x < 2 \\ \frac{1}{18}(3+2x) ; 2 \leq x \leq 4 \\ 0 ; x \geq 4 \end{cases}$

Soln Now, $\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{2} f(x)\,dx + \int_{2}^{4} f(x)\,dx + \int_{4}^{\infty} f(x)\,dx.$

$$= \int_{-\infty}^{2} 0 + \int_{2}^{4} \frac{1}{18}(3+2x)\,dx + \int_{4}^{\infty} 0\,dx.$$

$$= 0 + \int_{2}^{4} \frac{1}{18}(3+2x)\,dx + 0.$$

$$= \frac{1}{18}\left(x + x\frac{x^2}{x}\right)_2^4 \Rightarrow \frac{1}{18}\left(2 + 4 - 4 + 16\right.$$

$$= \frac{18}{18} = 1.$$

$\therefore$ The fun^n $f(x)$ is a valid density fun^n.

(3) If the probability density fun^n of R.V is given by

$$f(x) = \begin{cases} K(1-x^2) ; 0<x<1 \\ 0 ; \text{elsewhere} \end{cases}$$ find K and

i) $P(0.1<x<0.2)$; ii) $P(x>0.5)$

Solt Now $\int_{-\infty}^{0} f(x)dx + \int_{0}^{1} f(x)dx + \int_{1}^{\infty} f(x)dx = 1$

$$= \int_{-\infty}^{0} K(1-x^2)dx + \int_{0}^{1} K(1-x^2)dx + \int_{1}^{\infty} K(1-x^2)dx = 1$$

$$= 0 + \int_{0}^{1} K(1-x^2)dx + 0 = 1$$

$$= K\left(x - \frac{x^3}{3}\right)_0^1 = 1 \Rightarrow K\left(1-\frac{1}{3}\right) = 1$$

$$= K\left(\frac{2}{3}\right) = 1 \Rightarrow K = \frac{3}{2}$$ //

i) $P(0.1<x<0.2)$

$$= \int_{0.1}^{0.2} f(x)\,dx \Rightarrow \int_{0.1}^{0.2} \frac{3}{2}(1-x^2)\,dx$$

$$= \frac{3}{2}\left(x - \frac{x^3}{3}\right)_{0.1}^{0.2} =$$

$$= \frac{3}{2}\left[\left((0.2) - \frac{(0.2)^3}{3}\right) - \left(0.1 - \frac{(0.1)^3}{3}\right)\right]$$

$$= 0.1465$$

ii) $P(x>0.5)$

$$= P(0.5<x) \Rightarrow \int_{0.5}^{\infty} f(x)\,dx \Rightarrow \int_{0.5}^{\infty} K(1-x^2)\,dx$$

$$\Rightarrow \int_{0.5}^{\infty} \frac{3}{2}(1-x^2)\,dx \Rightarrow \frac{3}{2}\left(x - \frac{x^3}{3}\right)_{0.5}^{\infty} dx.$$

$$\frac{3}{2}\left(0.5 - \frac{(0.5)^3}{3}\right) + \left(1 - \frac{1}{3}\right) \Rightarrow$$

Cummulative distribution function :-

Suppose that, X is discrete R.V then the cummulative distributive function defined as

$$\Rightarrow f(x) = p(x \leq x) = \sum_{i=1}^{x} p(x)$$

if x is continous R.V then f(x) is

$$\Rightarrow f(x) = p(x \leq x) = \int_{-\infty}^{x} f(x)\,dx$$

NOTE :- 1) $F(-\infty) = 0$

2) $F(\infty) = 1$

3) $\frac{d}{dx}[F(x)] = f(x)$

4) $F(x) = \int_{-\infty}^{\infty} f(x)\,dx$

Mathematical Expectation :-

The Average process when apply to the random Vairable is called the Expectation and It is denoted by E(x) Read as Expected value of R.V of x. (or) mean value of x.

Defn Suppose 'x' is a Discrete R.V which takes on the values in a finite set $x_1, x_2, x_3 \cdots x_n$ with probabilities P(Xi) for $i = 1, 2, 3 \cdots n$, then the expectation E(x) of the discrete R.V 'x' is given by

$$E(x) = \text{mean} = \sum_{i=1}^{n} x_i\, p(x_i)$$

Provided this sereis is convergent.

⇒ If X is a continious R.V then

$$E(x) = \int_{-\infty}^{\infty} x f(x)\, dx \; ; \; -\infty < x < \infty.$$

NOTE (1) : $E(x^2) = \sum_{i=1}^{n} x_i^2 \, p(x_i)$; X is Discrete R.V

(2) $E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\, dx$; X is continous R.V

(3) $E[g(x)] = \sum_{i=1}^{n} g(x_i)\, p(x_i)$ if X is Discrete R.V

(4) $E[g(x)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx$ if X is a Cont. R.V

Properties :

Let X, Y are two R.Vairables and a, b are constants

then (i) $E(a) = a$

(ii) $E(ax) = a E(x)$

(iii) $E(ax+b) = a E(x) + b$

(iv) $E(ax \pm by) = a E(x) \pm b E(Y)$

(v) $E(x+a) = E(x) + a$

(vi) If X, Y, Z are independent R.V then

$$E(XYZ) = E(X) \cdot E(Y) \cdot E(Z)$$

(vii) $E(x) = m = \mu$.

Vairance :

If 'x' is R.V then the Vairance of 'x' is denoted by $V(x)$ (or) $\sigma_x^2 = E\left[\{x - E(x)\}^2\right]$

$$= E\left[(x-\mu)^2\right]$$

$$= E(x^2) - \mu^2$$

NOTE : 1) If X is D.R.V then

$$V(x) = E(x^2) - \mu^2$$

$$= E(x^2) - [E(x)]^2$$

$$= \sum_{i=1}^{n} x_i^2 \, p(x_i) - \left[\sum_{i=1}^{n} x_i \, p(x_i)\right]^2$$

2) If x is C.R.V then

$$V(x) = E(x^2) - [E(x)]^2$$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - \left[\int_{-\infty}^{\infty} x f(x)\,dx\right]^2$$

Standard Deviation :-

A positive square root of a vairable is called standard Deviation and it is denoted by 'σx'.

$$\boxed{S.D = \sigma x = \sqrt{V(x)}}$$

properties :-

Let x, y are two R.V and a, b are constants then

(i) $V(a+x) = V(x)$

(ii) $V(ax) = a^2 V(x)$

(iii) $V(a) = 0$

(iv) $V(ax+b) = a^2 V(x)$

(v) $V(ax \pm by) = a^2 V(x) \pm b^2 V(y)$

(vi) $V(x \pm y) = V(x) + V(y) \pm 2[E(xy) - E(x).E(y)]$

Here $E(XY) - E(X) \cdot E(Y)$ is called co-vairance of x and y.

1Q]:- The R.V 'x' as the following probability fun^n

X=x	-3	-2	-1	0	1	2	3
P(x)	K	0.1	K	0.2	2K	0.4	2K

Find (i) K (ii) Mean (iii) vairance.

Soln i) w.k.T total probability is unit (1) i.e

$$\sum_{i=1}^{7} p(x_i) = 1 \Rightarrow \sum_{-3}^{3} p(x) = 1$$

$$p(x=-3) + p(x=-2) + p(-1) + p(0) + p(1) + p(2) + p(3) = 1$$

$$K + 0.1 + K + 0.2 + 2K + 0.4 + 2K = 1$$

$$6K + 0.6 + 0.1 = 1 \Rightarrow 6K + 0.7 = 1$$

$$= K = \frac{1}{20}$$

ii) Mean $= E(x) = \sum_{i=1}^{7} x_i\, p(x_i)$

$$\mu = (-3)\,p(-3) + (-2)\,p(-2) + (-1)\,p(-1) + 0 \cdot p(0) + 1 \cdot p(1) + 2\,p(2) + 3\,p(3)$$

$$= (-3)\,K + (-2)(0.1) + (-1)(K) + 0 + (1)(2K) + 2(0.4) + 3(2K)$$

$$= (-3)\left(\frac{1}{20}\right) - 2(0.1) - \frac{1}{20} + 2\left(\frac{1}{20}\right) + 2(0.4) + 3\left(\frac{1}{20}\right)$$

μ = Mean = 0.8

iii) Vairance :

$$\boxed{V(x) = E(x^2) - \mu^2}$$

Now, $E(x^2) = \sum_{i=1}^{7} x_i^2\, p(x_i)$

$$= \sum_{-3}^{3} x^2\, p(x)$$

$$= (-3)^2 p(-3) + (-2)^2 p(-2) + (-1)^2 p(-1) + 0^2 p(0) + 1^2 p(1) + 2^2 p(2) + 3^2 p(3)$$

$$E(x^2) = 9p(-3) + 4p(-2) + p(-1) + 0 + p(1) + 4p(2) + 9p(3)$$

$$= 9(K) + 4(0.1) + K + 0 + 2K + 0.4 + 2K$$

$$E(x^2) = 3.5$$

$\therefore V(X) = E(X^2) - \mu^2$

$= 3.5 - (0.8)^2 = 2.86$

①

Q) A R.V 'X' as a following probability function.

X	1	2	3	4	5	6
P(X)	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

i) Find K ii) mean iii) variance.

i) Total probability is unity :

Soln i.e $\sum_{i=1}^{n} p(x) = 1$

$\sum_{i=1}^{6} P(X=x_i) = 1$

$\Rightarrow p(X=1) + p(X=2) + \cdots + p(X=6) = 1$

$2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$

$10K^2 + 8K - 1 = 0.$

$K = \frac{-4 \pm \sqrt{26}}{10} = 0.109 = 0.1$

$\boxed{K = 0.1}$

ii) Mean : $E(X) = \sum_{i=1}^{n} x_i p(x_i)$

$= \sum_{i=1}^{6} x_i p(x_i)$

$= (1)p(1) + 2p(2) + 3p(3) + \cdots + 6p(6)$

$= 1(2K) + 2(2K) + 3(3K) + 4(K^2) + 5(2K^2) + 6(7K^2 + K)$

$= 2K + 4K + 9K + 4K^2 + 10K^2 + 42K^2 + 6K.$

$= 2(0.1) + 4(0.1) + 9(0.1) + 4(0.1)^2 + 10(0.1)^2 + 42(0.1)^2 + 6(0.1)$

$\mu = E(X) = 2.66$

iii) Variance :- $\boxed{V(x) = E(x^2) - \mu^2}$

$$= \sum_{i=1}^{n} x_i^2 p(x_i) \Rightarrow \sum_{i=1}^{6} x_i^2 p(x_i)$$

$= 1\,p(1) + 4\,p(2) + 9p(3) + 16(p(4) + 25p(5) + 36\,p(6)$

$= 1(2k) + 4(2k) + 9(3k) + 16(k^2) + 25(2k^2) + 36(7k^2 + k)$

$= 2k + 8k + 27k + 16k^2 + 50k^2 + 252k^2 + 36k$

$= 2(0.1) + 8(0.1) + 27(0.1) + 16(0.1)^2 + 50(0.1)^2 + 252(0.1)^2 + 36(0.1)$

$\Rightarrow 10.48 - (2.66)^2 = 3.41$

10.48
2.66
7.82

2/ Q) A R.V 'x' has the following prob. distribution.

$X = x$:	1	2	3	4	5	6	7	8
$P(x)$:	k	2k	3k	4k	5k	6k	7k	8k

i) Find k ii) $P(X \leq 2)$ iii) $P(2 \leq X \leq 5)$

Sol:- i) Total probability is unity.

i.e $\sum_{i=1}^{n} P(x) = 1 \Rightarrow \sum_{i=1}^{8} P(x) = 1$.

$\Rightarrow k + 2k + 3k + 4k + 5k + 6k + 7k + 8k = 1$

$36k = 1 \Rightarrow k = \frac{1}{36}$

ii) $P(X \leq 2)$

$= P(X=1) + P(X=2) \Rightarrow k + 2k$

$\Rightarrow 3k = 36$

$\Rightarrow k = \frac{3}{36} = \frac{1}{12}$

iii) $P(2 \leq X \leq 5)$

$P(X=2) + P(X=3) + P(X=4) + P(X=5)$

$2k + 3k + 4k + 5k = 14k$

$\Rightarrow k = 14/36$

Q) Let 'x' denotes the minimum of the two numbers that appear a pair of square fair dice thrown on a determine

i) Discrete probability distrubiton.

ii) Expectation.

iii) Variance

iv) Standard deviation.

Sol)

$$S = \begin{Bmatrix} (1,1)\ (1,2)\ (1,3)\ (1,4)\ (1,5)\ (1,6) \\ (2,1)\ (2,2)\ (2,3)\ (2,4)\ (2,5)\ (2,6) \\ (3,1)\ (3,2)\ (3,3)\ (3,4)\ (3,5)\ (3,6) \\ (4,1)\ (4,2)\ (4,3)\ (4,4)\ (4,5)\ (4,6) \\ (5,1)\ (5,2)\ (5,3)\ (5,4)\ (5,5)\ (5,6) \\ (6,1)\ (6,2)\ (6,3)\ (6,4)\ (6,5)\ (6,6) \end{Bmatrix}$$

Given that $X(a,b) = \min\{a,b\}$

The Minimum no.s would be 1, 2, 3, 4, 5, 6

i,e

$$\begin{Bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 3 & 4 & 4 & 4 \\ 1 & 2 & 3 & 4 & 5 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{Bmatrix}$$

These are The favorable cases.

for min=1, the favorable cases are

$$P(X=1) = P\{(1,1)\ (1,2)\ (1,3)\ (1,4)\ (1,5)\ (1,6)\ (2,1)\ (3,1)\ (4,1)\ (5,1)\ (6,1)\}$$

$$= \frac{11}{36}$$

for min = 2; $P(X=2) \Rightarrow P\{(2,2)\ (2,3)\ (2,4)\ (2,5)\ (2,6)\ (3,2)\ (4,2)\ (5,2)\ (6,2)\}$

$$= 9/36$$

for min = 3

$$P(X=3) = \frac{7}{36}$$

for min = 4 $P(X=4) = \frac{5}{36}$

for min = 5 ; $P(X=5) = \frac{3}{36}$

for min 6 ; $P(X=6) = \frac{1}{36}$

∴ The prob. distribution fun^n for given R.V 'x' is

$X = x$	1	2	3	4	5	6	
$P(x=x)$	$11/36$	$9/36$	$7/36$	$5/36$	$3/36$	$1/36$	

ii) Expectation (mean) :

$$\sum_{i=1}^{?} x_i p(x_i) \Rightarrow \sum_{i=1}^{6} x_i p(x_i)$$

$$1(p(1) + 2p(2) + 3(p(3) + 4p(4) + 5P(5) + 6p(6)$$

$$= 1\left(\frac{11}{36}\right) + 2\left(\frac{9}{36}\right) + 3\left(\frac{7}{36}\right) + 4\left(\frac{5}{36}\right) + 5\left(\frac{3}{36}\right) + 6\left(\frac{1}{36}\right)$$

$$= \frac{11}{36} + \frac{18}{36} + \frac{21}{36} + \frac{20}{36} + \frac{15}{36} + \frac{6}{36}$$

$$= \frac{91}{36} = 2.5278$$

iii) Variance

$$V(x) = E(X^2) - \mu^2$$

$$= 1P(1) + 4p(2) + 9(p(3) + 16p(4) + 25p(5) + 36p(6)$$

$$= \frac{11}{36} + 4\left(\frac{9}{36}\right) + 9\left(\frac{7}{36}\right) + 16\left(\frac{5}{36}\right) + 25\left(\frac{3}{36}\right) + 36\left(\frac{1}{36}\right)$$

$$= \frac{11}{36} + \frac{36}{36} + \frac{63}{36} + \frac{80}{36} + \frac{75}{36} + \frac{36}{36}$$

$E(X^2) = 8.3611 = 8.39$,

$= 7.39 - (2.5278)^2$

$= 8.3611 - (2.5278)^2 = 1.9721$

iv) S.D $= \sqrt{V(x)} = \sqrt{1.9721} = 1.40431$

Q) A R.V 'X' a probability disturition as following table

X	0	1	2	3	4	5	6	7
P(x)	a	3a	5a	7a	9a	11a	13a	15a

i) Determine the value of 'a'.

ii) Find $P(X<3)$, $P(X \geq 3)$, $P(0<X<5)$

iii) Find the distribution fun^n F(x). (cumilative)

i) $a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$

$\Rightarrow 81a = 1 \Rightarrow a = \frac{1}{81}$

ii) $P(X<3) \Rightarrow P(X=0) + P(X=1) + P(X=2)$

$a + 3a + 5a = 9$

$9a = \frac{1}{81} \Rightarrow a = 9/81$

$P(X \geq 3) \Rightarrow P(X=3) + P(X=4) + P(X=5) + P(X=6) + P(X=7)$

$\Rightarrow 7a + 9a + 11a + 13a + 15a + 17a = 81$

$72a = 81$

$a = \frac{72}{81}$ (or)

$1 - P(X<3) \Rightarrow 1 - \frac{9}{81} = \frac{72}{81}$

$P(0<X<5)$

$P(X=1) + P(X=2) + P(X=3) + P(X=4)$

$3a + 5a + 7a + 9a = 81$

$24a = 81 \Rightarrow a = \frac{24}{81}$

(iii)

$X = x$	$F(x) = P(X \le x) = \sum_{i=0}^{x} P(x_i)$
$X = 0$	$F(x) = a = \frac{1}{81}$
$X = 1$	$F(1) = p(0) + p(1) = a + 3a = 4a = \frac{4}{81}$
$X = 2$	$F(2) = 9a \Rightarrow \frac{9}{81}$
$X = 3$	$F(3) = 16a = \frac{16}{81}$
$X = 4$	$F(4) = 25a = \frac{25}{81}$
$X = 5$	$F(5) = 36a \Rightarrow \frac{36}{81}$
$X = 6$	$F(6) = 49a \Rightarrow \frac{49}{81}$
$X = 7$	$F(7) = 64a \Rightarrow \frac{64}{81}$
$X = 8$	$F(8) = 81a \Rightarrow \frac{81}{81}$

$\Rightarrow$ (5) $x(a,b) = \max\{a,b\}$

6) Q $\Rightarrow$ Find the mean and variance of uniform probability distribution given by $f(x) = \frac{1}{n}$; $x = 1, 2, 3 \cdots n$.

X	1	2	3	4	------	n
$f(x)$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	------	$\frac{1}{n}$

Sol (i) Find mean

$$E(X) = \mu = \sum_{i=1}^{n} x_i p(x_i)$$

$$= p(x=1) + p(x=2) + p(x=3) + \cdots + p(x=n)$$

$$= 1\,p(x=1) + 2\,p(x=2) + \cdots + n\,p(x=n)$$

$$= \frac{1}{n} + 2\frac{1}{n} + 3\frac{1}{n} + \cdots + n\frac{1}{n}$$

$$\Rightarrow \frac{1}{n}\left(1 + 2 + 3 + \cdots + n\right)$$

$$\frac{1}{n}\left(\frac{n(n+1)}{2}\right) \Rightarrow \frac{n+1}{2}$$

ii) Variance $\Rightarrow V(x) = E(x^2) - \mu^2$

$$\Rightarrow 1p(1) + 4[p(2) + 9\,p(3) + 16\,p(4) + \cdots n^2 p(n)$$

$$\Rightarrow \frac{1}{n} + \frac{4}{n} + \frac{9}{n} + \frac{16}{n} + \cdots \frac{n^2}{n}$$

$$\frac{1}{n}\left[1^2 + 2^2 + 3^2 + 4^2 + \cdots n^2\right]$$

$$\frac{1}{n}\left[\frac{n(n+1)(2n+1)}{6}\right]$$

$$E(x^2) \Rightarrow \frac{(n+1)(2n+1)}{6}$$

$$\mu^2 = \frac{(n+1)^2}{4}$$

$$\frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$= (n+1)\left[\frac{2n+1}{6} - \frac{n+1}{4}\right]$$

$$= n+1\left[\frac{4(2n+1) - 6(n+1)}{24}\right]$$

$$= n+1\left[\frac{8n+4-6n+6}{24}\right]$$

$$= (n+1)\frac{(n-2)}{12} \Rightarrow \frac{n^2-1}{12}$$

7) A sample of four items is selected at random from a box, containing 12 items of which five are defective find the Expected no. E of defective items.

Soln Let 'x' denotes number of defective items.

No. of good items = 7.

No. of defective items = 5 and

$$n(S) = {}^{12}C_4$$

Now, $P(X=0) = \frac{{}^7C_4 \times {}^5C_0}{{}^{12}C_4} = \frac{7}{99}$

$P(X=1) = \frac{{}^7C_3 \times {}^5C_1}{{}^{12}C_4} = \frac{35}{99}$

$P(X=2) = \frac{{}^7C_2 \times {}^5C_2}{{}^{12}C_4} = \frac{42}{99}$

$P(X=3) = \frac{{}^7C_1 \times {}^5C_3}{{}^{12}C_4} = \frac{14}{99}$

$P(X=4) = \frac{{}^7C_0 \times {}^5C_4}{{}^{12}C_4} = \frac{1}{99}$

The probability dist^n table is.

X	0	1	2	3	4
P(X)	$\frac{7}{99}$	$\frac{35}{99}$	$\frac{42}{99}$	$\frac{14}{99}$	$\frac{1}{99}$

Mean $= E(X) = \sum_{i=0}^{4} x_i p(x_i)$

$= 0p(X=0) + 1p(X=1) + 2p(X=2) + 3p(X=3) + 4p(X=4)$

$= 0 \times \left(\frac{7}{99}\right) + 1\left(\frac{35}{99}\right) + 2\left(\frac{42}{99}\right) + 3\left(\frac{14}{99}\right) + 4\left(\frac{1}{99}\right)$

$= \frac{35 + 84 + 42 + 4}{99} = \frac{165}{99} = \frac{5}{3}$

Sm

6) Show that variance of R.V 'x' is given by

$\sigma^2 = E(X^2) - (E(X))^2$

$$V(x) = \sigma_x^2 = E[(x-\mu)^2]$$

$$E[x^2 + \mu^2 - 2x\mu]$$

$$E(x^2) + E(\mu^2) - 2E(x)E(\mu)$$

$$= E(x^2) + \mu^2 - 2\mu\mu \quad (\because E(a) = a)$$

$$= E(x^2) + \mu^2 - 2\mu^2$$

$$V(x) = E(x^2) - \mu^2$$

$$\sigma_x^2 = E(x^2) - [E(x)]^2$$

9) A variate (or) RV 'x' is

x	-3	6	9
P(x)	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

then find (i) $E(x)$ ii) $E(x^2)$ iii) $E[(2x+1)^2]$

Soln i) $E(x) = \sum_{i=0}^{n} x_i P(x_i) \Rightarrow \sum_{i=-3}^{9} x_i p(x_i)$

$$= -3\left(\frac{1}{6}\right) + 6\left(\frac{1}{2}\right) + 9\left(\frac{1}{3}\right)$$

$$= \frac{-1}{2} + 3 + 3 = \frac{-1+6+6}{2} = \frac{+11}{2}$$

ii) $E(x^2) = \sum_{i=0}^{n} x_i^2 p(x_i)$

$$= (-3)^2\left(\frac{1}{6}\right) + 36\left(\frac{1}{2}\right) + 81\left(\frac{1}{3}\right)$$

$$= \frac{9}{6} + \frac{36}{2} + \frac{81}{3} = \frac{3}{2} + 18 + 27$$

$$= \frac{3+36+54}{2} = \frac{93}{2}$$

iii) $E[(2x+1)^2]$

$$E[4x^2 + 1 + 4x]$$

$$4E(x^2) + E(1) + 4E(x)$$

$$= 4\left(\frac{93}{2}\right) + 1 + 4\left(\frac{11}{2}\right) = 186 + 1 + 22$$

$$= 209 \text{ //}$$

H/W:- Let 'x' denotes the minimum (maximum) of two number that appear a pair of square fair dice thrown on a determine.

i) Discrete probability distribution.

ii) Expectation

iii) Variance

iv) Standard deviation.

Solr Let $X = \max(a, b)$

$n(S) = 36$

$X = 1, 2, 3, 4, 5, 6.$

Discrete prob. distribution is

for max = 1, the favorable cases are

$P(X=1) = (1,1) \Rightarrow \frac{1}{36}$

for max = 2, the favorable cases are

$P(X=2) = (1,2)(2,1)(2,2) = \frac{3}{36}$

for max = 3, the favorable cases are

$P(X=3) = (3,1)(3,2)(3,3)(2,3)(1,3)$

$= \frac{5}{36}$

for max = 4; the favorable cases are

$P(X=4) = (4,1)(4,2)(4,3)(4,4)(3,4)(2,4)(1,4)$

$P(X) = \frac{7}{36}$

$P(X=5)$ = the favorable cases are $\frac{9}{36}$

$P(X=6)$ ⇒ The favorable cases are $\frac{11}{36}$

X	1	2	3	4	5	6
P(X)	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

ii) Expectation value $E(x)$

$$E(X) = \sum x_i p(x_i)$$

$$= \frac{1(1) + 2(3) + 3(5) + 4(7) + 5(9) + 6(11)}{36}$$

$$E(x) = \frac{161}{36} = 4.4722$$

iii) Variance :-

$$E(x^2) = \sum x^2 p(x)$$

$$= \frac{1^2(1) + 2^2(3) + 3^2(5) + 4^2(7) + 5^2(9) + 6^2(11)}{36}$$

$$E(x^2) = \frac{791}{36}.$$

$$V(x) = E(x^2) - \mu^2$$

$$= \frac{791}{36} - \left(\frac{161}{36}\right)^2 = \frac{2555}{1296} \approx 1.9715$$

$$V(x) = 2.$$

iv) Standard deviation :-

$$\sigma = \sqrt{V(x)} = \sqrt{\frac{2555}{1296}} = 1.404$$

* Problems on continious Random Variable :-

*** 1) The probability density $f(x)$ of continious RV is by $f(x) = ce^{-|x|}$; $-\infty < x < \infty$. then show that $c = \frac{1}{2}$ and find the mean and variance of the distribution also find the probability of the variant lies b/n 0 & 4

Sol:- Given that, $f(x) = ce^{-|x|}$; $-\infty < x < \infty$

we have to find (i) c value
(ii) mean
(iii) variance
(iv) $P(0 < x < 4) = ?$

(i) W.K.T $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\int_{-\infty}^{\infty} ce^{-|x|}\,dx = 1$

$= c\,2\int_{0}^{\infty} e^{-|x|}dx = 1$

$2c\int_{0}^{\infty} e^{-x}\,dx = 1$

$2c\left(\frac{e^{-x}}{-1}\right)_0^{\infty} = 1$

$-2c\,[e^{-\infty} - e^{+0}) = 1$

$-2c\,[0-1] = 1 \Rightarrow 2c = 1 \Rightarrow c = 1/2$ hence proved //

$\because \int_{-a}^{a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx & f(x) \to \text{even} \\ 0 & ; f(x) \text{ odd} \end{cases}$

$f(x) = x^2$

$f(-x) = f(x) \Rightarrow$ even

$f(-x) = -f(x) \Rightarrow$ odd

ii) Mean $E(x)$:-

$E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$

$= \int_{-\infty}^{\infty} x \cdot c \cdot e^{-|x|}\,dx$

$= \frac{1}{2}\int_{-\infty}^{\infty} x\,e^{-|x|}\,dx$ because → odd funtion then it is '0'.

$= 0$

iii) Variance $V(x)$:-

$V(x) = E(x^2) - [E(x)]^2 = E(x^2) - \mu^2$

$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - \left[\int_{-\infty}^{\infty} x f(x)\,dx\right]^2$

$= \int_{-\infty}^{\infty} x^2 \cdot c \cdot e^{-|x|}\,dx - \left[\int_{-\infty}^{\infty} x\,e^{-|x|}\,dx\right]^2$

$= \frac{1}{2}\int_{-\infty}^{\infty} x^2 e^{-|x|}\,dx - 0.$

$V(x) = \frac{1}{2}\int_{-\infty}^{\infty} x^2 e^{-|x|}\,dx$

$= \int_{0}^{\infty} x^2 e^{-x}\,dx$

$\left(\because \int x^2 e^{ax}dx = \frac{x^2 e^{ax}}{a} - \frac{2x e^{ax}}{a^2} + \frac{2e^{ax}}{a^3}\right.$

$$\because \int x^2 e^{-ax}\,dx = \frac{x^2 e^{-ax}}{(-a)} - \frac{2x e^{-ax}}{(-a)^2} + \frac{2e^{-ax}}{(-a)^3}$$

$$= \left[\frac{x^2 e^{-x}}{(-1)^1} - \frac{2x e^{-x}}{(-1)^2} + \frac{2e^{-x}}{(-1)^3}\right]_0^\infty$$

$$= (0) - \left(0 - 0 + \frac{2e^0}{(-1)}\right)$$

$$= 2 \text{ //}$$

iv) $P(0<x<4) = \int_0^4 f(x)\,dx$

$$= \int_0^4 c\, e^{-|x|}\,dx$$

$$= \frac{1}{2}\int_0^4 e^{-x}\,dx \Rightarrow \frac{1}{2}\left(\frac{e^{-x}}{-1}\right)_0^4$$

$$= \frac{-1}{2}\left(e^{-4} - e^0\right)$$

$$= 0.4908 \text{ //}$$

2) A Continous RV has a probability density functio

$$f(x) = \begin{cases} kx\, e^{-\lambda x} ; & x \geq 0, \lambda > 0 \\ 0 ; & \text{otherwise} \end{cases}$$

Determine (i) K ii) mean iii) Variance.

Soln W.K.T $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$= \int_{-\infty}^{0} f(x)\,dx + \int_0^\infty f(x)\,dx = 1$$

$$= 0 + \int_0^\infty kx\, e^{-\lambda x}\,dx = 1$$

$$= k\int_0^\infty x\, e^{-\lambda x}\,dx = 1$$

$$k\left[\frac{x e^{-\lambda x}}{-\lambda} - \frac{1 \cdot e^{-\lambda x}}{(-\lambda)^2}\right]_0^\infty = 1$$

$$k\left[0 - \frac{e^{-(0)}}{(-\lambda)^2} - 0\right] = 1$$

$$k\left[-\lambda^2\right] = 1 \Rightarrow k = \lambda^2$$

ii) Mean $E(x)$:

$$E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$= \int_{0}^{\infty} x \cdot kx e^{-\lambda x}\,dx$$

$$= \lambda^2 \int_{0}^{\infty} x^2 \cdot e^{-\lambda x}\,dx$$

$$= \lambda^2 \left[\frac{x^2 e^{-\lambda x}}{-\lambda} - \frac{2x e^{-\lambda x}}{(-\lambda)^2} + \frac{2e^{-\lambda x}}{(-\lambda)^3} \right]_0^{\infty}$$

$$= \lambda^2 \left[\frac{0 e^{-\lambda(0)}}{-\lambda} - \frac{2(0)e^{-\lambda(0)}}{(-\lambda)^2} + \frac{2e^{-\lambda(0)}}{(-\lambda)^3} + \frac{2}{\lambda^3} \right]$$

$$\lambda^2 \left[0 + \frac{2}{\lambda^3} \right]$$

$$E(x) = 2/\lambda$$

iii) Variance

$$V(x) = E(x^2) - [E(x)]^2 = E(x^2) - \mu^2$$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - \left(\frac{2}{\lambda}\right)^2$$

$$= \int_{-\infty}^{0} x^2 f(x)\,dx - \left(\frac{2}{\lambda}\right)^2 + \int_{0}^{\infty} x^2 f(x)\,dx - \left(\frac{2}{\lambda}\right)^2$$

formula: $\int x^n e^{-\lambda x}\,dx$

$$\Rightarrow 0 + \int_{0}^{\infty} x^2 k e^{-\lambda x}\,dx - \frac{4}{\lambda^2}$$

$$\frac{x^3 e^{-\lambda x}}{-\lambda} - \frac{3x^{n-1} e^{-\lambda x}}{(-\lambda)^2} + \frac{n(n-1)x^{n-2} e^{-\lambda x}}{(-\lambda)^3} \cdots$$

$$= \lambda^2 \int_{0}^{\infty} x^3 e^{-\lambda x}\,dx - \frac{4}{\lambda^2}$$

$$= \lambda^2 \left(\frac{x^3 e^{-\lambda x}}{-\lambda} - \frac{3x^2 e^{-\lambda x}}{(-\lambda)^2} + \frac{6x e^{-\lambda x}}{(-\lambda)^3} - \frac{6e^{-\lambda x}}{(-\lambda)^4} \right)_0^{\infty} - \frac{4}{\lambda^2}$$

$$= \lambda^2 \left[0 - \left[0 - 0 + 0 - \frac{6e^{-\lambda(0)}}{(-\lambda)^4} \right] \right] - \frac{4}{\lambda^2} \Rightarrow \lambda^2 \left[0 + \frac{6}{\lambda^4} \right] - \frac{4}{\lambda^2}$$

$$= \frac{2}{\lambda^2}$$

3) A continous R.V 'x' is defined by $f(x) = \frac{1}{16}$

$$f(x) = \begin{cases} \frac{1}{16}(3+x)^2 ; & -3 \le x \le -1 \\ \frac{1}{16}(6 - 2x^2) ; & -1 \le x \le 1 \\ \frac{1}{16}(3 - x^2) , & 1 \le x \le 3 \end{cases}$$

0, Evaluate else where.

⇒ Verify that $f(x)$ is density function? Also find mean of 'X'

Sol:

Now $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{-3} f(x)dx + \int_{-3}^{1} f(x)dx + \int_{-1}^{1} f(x)dx + \int_{1}^{3} f(x)dx + \int_{+3}^{\infty} f(x)dx$

$= 0 + \int_{-3}^{1} \frac{(3+x)^2}{16}dx + \int_{-1}^{1} \frac{6-2x^2}{16}dx + \int_{1}^{3} \frac{(3-x)^2}{16}dx + 0$

$= \frac{1}{16}\left[\int_{-3}^{-1}(3+x)^2dx + \int_{-1}^{1}(6-2x^2)dx + \int_{1}^{3}(3-x)^2dx\right]$

$= \frac{1}{16}\left[\int_{-3}^{-1}(x^2+6x+9)dx + \left\{\left(6x-\frac{2x^3}{3}\right)_{-1}^{1} + \int_{1}^{3}(x^2-6x+9)dx\right.\right.$

$= \frac{1}{16}\left[\frac{x^3}{3}+3x^2+9x\right]_{-3}^{-1} + \left(6x-\frac{2x^3}{3}\right)_{-1}^{1} + \left(\frac{x^3}{3}-3x^2+9x\right)_{1}^{3}$

$\frac{-27}{3}+27-27-6+\frac{2}{3}+\frac{3^3}{1}+27+9$

$= \frac{1}{16}\times\frac{48}{3} = \frac{16}{16} = 1$

Mean: $E(x) = \int_{-\infty}^{\infty} x f(x)dx$

$= \frac{1}{16}\left[\int_{-3}^{-1} x(3+x)^2dx + \int_{-1}^{1} x(6-2x^2)dx + \int_{1}^{3} x(3-x)^2dx\right]$

$= \frac{1}{16}\int_{-3}^{-1}(x^2+6x+9)dx + \left\{\left[3x^2-\frac{x^4}{2}\right]_{-1}^{1} + \int_{1}^{3}\left[x^2-6x+9\right]\right.$

$= \frac{1}{16}\left[\frac{x^4}{4}+2x^3+\frac{9}{2}x^2\right]_{-3}^{1} + \left[3x^2-\frac{x^4}{2}\right]_{-1}^{1} + \left[\frac{x^4}{4}-2x^3+\frac{9}{2}x^2\right]_{1}^{3}$

$= \frac{1}{16}\cdot\frac{81-216+162}{4} + \frac{3}{1}-\frac{1}{2}+3-\frac{1}{2}+\frac{81}{4}-54+\frac{81}{2}$

$= \frac{1}{16}(-4+0+4) = \frac{1}{16}(0) = 0$

$E(x) = 0$

A continious Random Variable 'x' have $F(x) = \begin{cases} 0 & \text{if } x \le 1 \\ k(x-1)^4, & \text{if } 1 \le x \le 3 \\ 0; & \text{if } x > 3 \end{cases}$

find i) $f(x)$ ii) k iii) Mean iv) Variance.

Sol: W.K.T the relation b/n cummilative distribution function $F(x)$ and probability distribution function $f(x)$, $\frac{d}{dx} F(x) = f(x)$

i) $\frac{d}{dx}[F(x)] = \begin{cases} 0; & x \le 1 \\ 4k(x-1)^3; & 1 \le x \le 3 \\ 0; & \text{if } x > 3 \end{cases}$

$f(x) = \begin{cases} 4k(x-1)^3; & 1 \le x \le 3 \\ 0; & \text{otherwise.} \end{cases}$

ii) WKT $\Rightarrow \int_{-\infty}^{\infty} f(x)\,dx = 1$

$= \int_{-\infty}^{1} f(x)\,dx + \int_{1}^{3} f(x)\,dx + \int_{3}^{\infty} f(x)\,dx = 1$

$= \int_{-\infty}^{1} 0\,dx + \int_{1}^{3} 4k(x-1)^3\,dx + \int_{3}^{\infty} 0 = 1$

$0 + 4k\left[\frac{(x-1)^4}{4}\right]_1^3 = 1$

$\Rightarrow 4k\left[\frac{(3-1)^4}{4}\right]_1^3 - \left[\frac{1-1}{4}\right]_1^3 = 1$

$\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1}$

$4k\left[\frac{16}{4}\right]^3 = 1$

$\boxed{k = \frac{1}{16}}$

iii) Mean $E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$.

$= \int_{1}^{3} x\, 4k(x-1)^3\,dx$.

$= 4\,\frac{1}{16} \int_{1}^{3} x(x-1)^3\,dx$

$(a-b)^3 = a^3 + b^3 + 3ab^2 + b^2$

$= \frac{1}{4}\left[x\left[\frac{x^4}{4} + 1 - 3x + 3x\right]_1^3\right]$

$$\mu = \frac{1}{4}\int_1^3 x^4 - 3x^3 + 3x^2 - x\,dx$$

$$= \frac{1}{4}\left[\frac{x^5}{5} - \frac{3x^4}{4} + \frac{3x^3}{3} - \frac{x^2}{2}\right]_1^3$$

$$= \frac{1}{4}\left[\frac{35}{5} - 3\frac{(3)^4}{4} + (3)^3 - \frac{(3)^2}{2} - \left(\frac{1}{5} - \frac{3}{4} + 1 - \frac{1}{2}\right)\right]$$

$$= \frac{1}{4}\left[\frac{243}{5} - \frac{243}{4} + 27 - \frac{9}{2} - \left(4 - \frac{15}{20} + \frac{2-1}{2}\right)\right]$$

$$= \frac{1}{4}\left[\frac{-243}{20} + \frac{54-9}{2} - \left(\frac{-11}{20} + \frac{1}{2}\right)\right]$$

$$= \frac{1}{4}\left[\frac{-243}{20} + \frac{45}{2} - \left(\frac{1}{20}\right)\right)$$

$$\mu = \frac{1}{4}\left[\frac{207}{20} + \frac{1}{20}\right] = \left(\frac{52}{5}\right)\frac{1}{4} = \frac{13}{5}$$

$$\boxed{\mu = \frac{13}{5}}$$

iv) Variance : $V(x) = E(x^2) - [E(x)]^2$

$$= \frac{1}{4}\int_1^3 x^2[x-1]^3\,dx - \left(\frac{13}{5}\right)^2$$

$$= \frac{1}{4}\int_1^3 x^2\left[x^3 - 3x^2 + 3x - 1\right] - \frac{169}{25}$$

$$= \frac{1}{4}\left[\frac{x^6}{6} - \frac{3x^5}{5} + \frac{3x^4}{4} - \frac{x^3}{3}\right]_1^3 - \frac{169}{25}$$

$$= \frac{1}{4}\left[\frac{3^6}{6} - 3\frac{(3)^2}{5} + \frac{3(3)^4}{4} - \frac{(3)^3}{3} - \left(\frac{1}{6} - \frac{3}{5} + \frac{3}{4} - \frac{4}{3}\right)\right.$$

$$= \frac{1}{4}\left[\frac{726}{6} - \frac{27}{5} - \frac{243}{4} - \frac{27}{3} - \left(\frac{1}{6} - \frac{3}{5} + \frac{3}{4} - \frac{1}{3}\right)\right]$$

$$V(x) = \frac{1}{4}\left[\frac{412}{15}\right] = \frac{103}{15} - \frac{169}{25}$$

$$\boxed{V(x) = \frac{8}{75}}$$

Imp*

5) For the continous probability fun^n $f(x) = kx^2e^{-x}$ when $x \geq 0$ find i) k ii) Mean iii) Variance

Sol^n $f(x) = kx^2e^{-x}$ $(x \geq 0)$

$$\int_{-\infty}^{0} f(x)\,dx + \int_0^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{0} kx^2e^{-x} + \int_0^{\infty} kx^2e^{-x} = 1$$

$$0 + \int_0^\infty k x^2 e^{-x} = 1$$

$$= k\left[\frac{x^3}{3} e^{-x}\right]_0^\infty = 1$$

$$= k\left[\frac{x^2 e^{-x}}{-1} + \frac{2x e^{-x}}{1^2} + \frac{2e^{-x}}{(-1)^3}\right]_0^\infty = 1$$

$$k\left[\frac{-x^2 e^{-x}}{+1} - 2x e^{-x} + (-2e^{-x})\right]_0^\infty = 1$$

$$k[0 - (0 - 0 - 2e^0)] = 1$$

$$k[0 - 0 - (2(1))] = 1 \Rightarrow k(2) = 1$$

$$\boxed{k = 1/2}$$

ii) Mean: $E(x) = \int_{-\infty}^{\infty} x f(x)\, dx$

$$= \int_0^\infty x \, k x^2 e^{-x}$$

$$= \frac{1}{2} \int_0^\infty x^3 e^{-x}$$

$$= \frac{1}{2}\left[\frac{x^3 e^{-x}}{(-1)} - \frac{3x^2 e^{-x}}{(-1)^2} + \frac{6e^{-x}}{(-1)^3} - \frac{6e^{-x}}{(-1)^4}\right]_0^\infty$$

$$= \frac{1}{2}\left[-x^3 e^{-x} + 3x^2 e^{x} - 6e^{-x} x \cdot - 6e^{-x}\right]_0^\infty$$

$$= \frac{1}{2}\left[0 - (0 + 0 + 0 - 6(e^{-0})\right]$$

$$= \frac{1}{2}[0 + 6(1)] = \frac{1}{2}(6)^3 = 3.$$

iii) Variance $V(x)$:

$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\, dx.$$

$$= \int_{-\infty}^{\infty} x^2 \left[\frac{1}{2} x^2 e^{-x}\right] dx$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} x^4 e^{-x}\, dx$$

$$= \frac{1}{2}\left[\frac{x^4 e^{-x}}{(-1)} - \frac{4x^3 e^{-x}}{(-1)^2} + \frac{12 x^2 e^{-x}}{(-1)^3} - \frac{24 x e^{-x}}{(-1)^4} + \frac{24 e^{-x}}{(-1)^5}\right]_0^\infty$$

$E(x^2) = \frac{1}{2}(24) = 12$

$V(x) = E(x^2) - [E(x)]^2 = 12 - (3)^2 = 12 - 9 = 3$

$V(x) = 3$,

6) If 'x' is a continious random variable distrubition where $y = ax+b$, prove that $E(Y) = aE(x)+b$ and $V(Y) = a^2 V(x)$ where 'V' stands a, b

Soln Given that $y = ax+b$

$$E(Y) = E(ax+b)$$

$$= \int_{-\infty}^{\infty} (ax+b) f(x)\, dx$$

$$= \int_{-\infty}^{\infty} ax f(x)\, dx + \int_{-\infty}^{\infty} b f(x)\, dx$$

$$\left(\because E(x) = \int_{-\infty}^{\infty} x f(x)\, dx\right)$$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\, dx$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx.$$

$$= a\int_{-\infty}^{\infty} x f(x)\, dx + b\int_{-\infty}^{\infty} f(x)\, dx$$

$$= aE(x) + b(1)$$

$$E(Y) = aE(x) + b \quad \text{---} (1)$$

(ii) from (i) we have $E(Y) = aE(x)+b$

and $y = ax+b$

$\therefore y - E(Y) = ax + b - aE(x) - b$

$$y - E(Y) = a[x - E(x)]$$

$$[y - E(Y)]^2 = a^2 [x - E(x)]^2$$

Taking expected value on both sides, we have

$$E\left[\{y - E(Y)\}^2\right] = a^2 E\left[\{x - E(x)\}^2\right]$$

$$V(Y) = a^2 V(x) \qquad \left(\because V(x) = E(x^2) - \mu^2\right.$$

7) If 'x' is a continous R.V and 'k' is a constant then Show that (i) $V(x+K) = V(x)$

(ii) $V(kx) = k^2 V(x)$

(i) Soln WKT $V(x) = E(x^2) - [E(x)]^2$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - \left[\int_{-\infty}^{\infty} x f(x)\,dx\right]^2$$

Now, $V(x+K) = E[(x+K)^2] - [E(x+k)]^2$

$$= \int_{-\infty}^{\infty} (x+K)^2 f(x)\,dx - \left[\int_{-\infty}^{\infty} (x+k) f(x)\,dx\right]^2$$

$$= \int_{-\infty}^{\infty} (x^2 + k^2 + 2kx) f(x)\,dx - \left[\int_{-\infty}^{\infty} x f(x)\,dx + k\int_{-\infty}^{\infty} f(x)\,dx\right]^2$$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx + k^2 \int_{-\infty}^{\infty} f(x)\,dx + 2k \int_{-\infty}^{\infty} x f(x)\,dx - [E(x) + k]^2$$

$$= E(x^2) + k^2(1) + 2k\,E(x) - [E(x)]^2 - k^2 - 2k\,E(x)$$

$$= E(x^2) - [E(x)]^2$$

$$= V(x)$$

(ii) $V(kx) = k^2 V(x)$

$$V(kx) = E[(kx)^2] - E[(kx)^2]^2$$

$$= \int_{-\infty}^{\infty} (kx)^2 f(x)\,dx - \left[\int_{-\infty}^{\infty} (kx) f(x)\,dx\right]^2$$

$$= \int_{-\infty}^{\infty} k^2 x^2 f(x)\,dx - \int_{-\infty}^{\infty} k^2 x^2 f(x^2)\,dx$$

$$[E(x)]^2$$

$$k^2 E(x^2) - [k E(x)]^2$$

$$k^2 E(x^2) - k^2 [E(x)]^2$$

$$V(kx) = k^2 [E(x^2) - [E(x)]^2]$$

$$k^2 V(x)$$

8) If the probability density fun^n of RV is given by

$$f(x) = \begin{cases} 2kxe^{-x^2}; & x > 0 \\ 0; & x \le 0 \end{cases}$$

find i) K

ii) The distribution fun^n. i.e. $F(x)$

Sol:- i) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{\infty} f(x)\,dx = 1$

$0 + \int_{0}^{\infty} 2kxe^{-x^2}\,dx = 1$

$\therefore k\int_{0}^{\infty} e^{-t}\,dx = 1$

$k\left[e^{-t}\right]_0^{\infty} = 1$

$-k\left[e^{\infty} - e^{0}\right] = 1$

$-k(-1) = 1$

$\boxed{k = 1}$

$\because x^2 = t$

$2x = \frac{d}{dt}$

$x = \infty, t$

$t = \infty;\ x = 0$

$t = 0$

ii) $\frac{dF(x)}{dx} = f(x)$

$F(x) = \int_{-\infty}^{x} f(x)$

$F(x) = 0$

$F(x) = \int_{0}^{x} 2kxe^{-x^2}\,dt \quad (k=1)$

$= \int_{0}^{x} 2xe^{-x^2}\,dx$

$= \int_{0}^{t^2} dt\, e^{-t}(0)$

$= \left[(-e^{-t})\right]_0^{t^2}$

$= -\left[e^{-t^2} - e^{0}\right]$

$= -\left[e^{-t^2} - 1\right]$

$1 - e^{-t^2}$

$1 - e^{-x^2}$

$\because t = x^2$

$x^2 = t$

$2x = \frac{dt}{dx}$

$2x\,dx = dt$

$\because t = x^2$

$x = 0$

$t = 0.$

1) The prob. density function of 'x' is given by

$$f(x) = \begin{cases} \frac{x}{2} ; & 0 < x \le 1 \\ \frac{1}{2} ; & 1 < x \le 2 \\ \frac{(3-x)}{2} ; & 2 < x \le 3 \\ 0 ; & \text{elsewhere} \end{cases}$$

find the expected value of $\phi(x)$. where $\phi(x) = x^2 - 5x + 3$

soln $\phi = x^2 - 5x + 3$

Expected value = ?

$$\int_{-\infty}^{\infty} x f(x)\, dx.$$

$$= \int_{-\infty}^{\infty} \phi(x) f(x) dx = \int_{\infty}^{\infty} \phi(x) F(x) dx + \int_{0}^{1} \phi(x) F(x) dx + \int_{1}^{2} \phi f(x) dx + \int_{2}^{3} \phi f(x)\, dx + \int_{3}^{\infty} \phi f(x)\, dx.$$

$$= 0 + \int_0^1 (x^2 - 5x + 3)\frac{x}{2} dx + \int_1^2 (x^2 - 5x + 3)\frac{1}{2} + \int_2^3 \frac{(3-x)}{2}(x^2 - 5x + 3)$$

$$= \frac{1}{2}\int_0^1 x^3 - 5x^2 + 3x\, dx + \frac{1}{2}\int_1^2 x^2 - 5x + 3\, dx + \frac{1}{2}\int_2^3 (3x^2 - 15x + 9 - 15x + 9 - x^3 - 5x^2 - 3x)\, dx.$$

$$= \frac{1}{2}\left[\frac{x^4}{4} - \frac{5x^3}{3} + 3\frac{x^2}{2}\right]_0^1 + \frac{1}{2}\left[\frac{x^3}{3} - 5\frac{x^2}{2} + 3x\right]_1^2 + \frac{1}{2}\left[\frac{3x^3}{3} - \frac{15x^2}{2} + 9x - \frac{x^4}{4} - \frac{5x^3}{3} - \frac{3x^2}{2}\right]_2^3$$

$$= \frac{1}{2}\left[\frac{1}{4} - \frac{5}{3} + \frac{3}{2}\right] + \frac{1}{2}\left[\frac{(2)^3}{3} - \frac{5(2)^2}{2} + 3(2) - \left(\frac{1}{3} - \frac{5}{2} + 3\right)\right]$$

$$= \frac{-3}{8} + \frac{26}{6} + \frac{1}{2}\left[-92 - \left(-\frac{79}{3}\right)\right].$$

$$= -\frac{3}{8} + \frac{26}{6} - \frac{92}{2} + \frac{79}{6} = -\frac{231}{84}$$

10) If 'x' is a continous Random Variable then the distribution function is $f(x) = \begin{cases} x; & 0 < x \le 1 \\ 2-x; & 1 \le x \le 2 \\ 0; & \text{otherwise.} \end{cases}$

find $E(25x^2 + 30x - 5)$

Sol: $E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$

$$\int_{-\infty}^{0} x f(x)\,dx + \int_{0}^{1} x f(x)\,dx + \int_{1}^{2} x f(x)\,dx + \int_{2}^{\infty} x f(x)\,dx$$

$$= 0 + \int_0^1 x(x)\,dx + \int_1^2 x(2-x)\,dx + 0$$

$$= \int_0^1 (x^2)\,dx + \int_1^2 (2x - x^2)\,dx$$

$$= \left(\frac{x^3}{3}\right)_0^1 + \left(2\frac{x^2}{2} - \frac{x^3}{3}\right)_1^2$$

$$\frac{1}{3} + \frac{2}{1} - \frac{8}{3} + \frac{1}{1} + \frac{1}{3} = \frac{1+6-8+3+1}{3}$$

$$E(x) = 3/3 = 1$$

$$\Rightarrow E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

$$= \int_{-\infty}^{0} x^2 f(x)\,dx + \int_0^1 x^2 f(x)\,dx + \int_1^2 x^2 f(x)\,dx + \int_2^{\infty} x^2 f(x)\,dx$$

$$= 0 + \int_0^1 x^2(x)\,dx + \int_1^2 x^2(2-x)\,dx + 0$$

$$= \int_0^1 x^3\,dx + \int_1^2 2x^2 - x^3\,dx$$

$$= \left[\frac{x^4}{4}\right]_0^1 + \left[2\frac{x^3}{3} - \frac{x^4}{4}\right]_1^2$$

$$= \frac{1}{4} + 2\frac{8}{3} - \frac{16}{4} + 2\frac{1}{3} - \frac{1}{4}$$

$$= \frac{1}{4} + \frac{16}{3} - \frac{16}{4} + \frac{2}{3} - \frac{1}{4}$$

$$= \frac{3 + 64 - 48 + 8 - 3}{12} = \frac{24}{12}$$

Now; $E[25x^2+30x-5]$

$= 25\,E(x^2) + 30\,E(x) - 5$

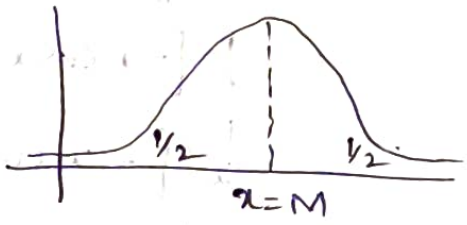
$= 25\left(\frac{5}{6}\right) + 30(1) - 5$

MEDIAN :-

Median is the point that divides Entire distribution into two equal parts.

$$\int_{-\infty}^{M} f(x)\,dx = \frac{1}{2} \quad \&$$

$$\int_{M}^{\infty} f(x)\,dx = \frac{1}{2}$$



Note:- If 'x' is denotes from a to b and M is median then $\int_{a}^{M} f(x)\,dx = \frac{1}{2}$ and $\int_{M}^{b} f(x)\,dx = \frac{1}{2}$

MODE :-

Mode is the value of 'x' for which $f(x)$ has maximum

$f'(x) = 0 \Rightarrow x = a, b$

$f''(x) \Rightarrow f''(a) < 0$

$f(x)$ has maximum at $x = a$

$f''(b) > 0$

$\Rightarrow f(x)$ has min at $x = b$

Mean Deviation :-

Mean deviation about the mean 'μ' is given by

$$\int_{-\infty}^{\infty} |x-\mu|\, f(x)\,dx.$$

1) ~~Find~~ prob. density function of a R.V 'x' is given

$$f(x) = \begin{cases} \frac{1}{2}\sin x\,; & 0 \le x \le \pi \\ 0; & \text{otherwise} \end{cases}$$

find mean, median, Mode, also find the prob. b/n 0 & $\frac{\pi}{2}$.

i) Mean :-

W.K.T $E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$.

$= \int_0^{\pi} x \frac{1}{2} \sin x\,dx$

$E(X) = \mu = \frac{1}{2} \int_0^{\pi} x \sin x\,dx$

$\mu = \frac{1}{2}\left[x \int \sin x\,dx - \int \frac{d}{dx}(x) \sin x\,dx\right]_0^{\pi}$

$\frac{1}{2}\left[-x\cos x - \int -\cos x\,dx\right]_0^{\pi}$

$= \frac{1}{2}\left[-x\cos x + \sin x\right]_0^{\pi}$

$\frac{1}{2}\left[-\pi\cos\pi + \sin\pi - (-0\cos(0) + 0)\right)$

$\frac{1}{2}\left(-\pi(-1) + 0 - 0\right) = \pi/2$ //

ii) Median :- $\int_{-\infty}^{M} f(x)\,dx = \frac{1}{2}$

$\int_0^M \frac{1}{2} \sin x\,dx = \frac{1}{2} \Rightarrow \int_0^M \sin x\,dx = 1$

$\left[-\cos x\right]_0^M = 1$

$-[\cos M - \cos 0] = 1$

$\cos M = 0 \Rightarrow M = \frac{\pi}{2}$

iii) Mode :- Now, $f(x) = \frac{1}{2}\sin x$

$f'(x) = 0$

$\Rightarrow \frac{1}{2}\cos x = 0$

$\cos x = 0$

$x = \pi/2$

$f''(x) = -\frac{1}{2}\sin x$

$\therefore f''\left(\frac{\pi}{2}\right) = \frac{-1}{2}\sin\frac{\pi}{2}$

$f''\left(\frac{\pi}{2}\right) < 0$ then at $x = \frac{\pi}{2}$,

$\therefore f(x)$ has max. value at $x = \frac{\pi}{2}$

* There are two types of prob. distribution — discrete prob. dist', continous prob.

1) DISCRETE PROB. DISTRIBUTION :-

There are mainly '3' Types.

(a) Binomial distribution
(b) possion distribution
(c) Geometrical distribution.

2) Continous prob. Distribution :-

there are '4' types.

(a) Normal distribution
(b) student t-distribution
(c) chi-square distribution
(d) F-distribution.

1) Discrete prob. distribution :-

a) Binomial distribution :-

A Random variable 'x' has a binomial distribution if it assume only non-negitive values and probability distribution that it given by'

$$P(X=x) = \begin{cases} {}^{n}C_x\, p^x q^{n-x} ; & x = 0, 1, 2, 3 \text{---} n \\ 0 ; & \text{otherwise} \end{cases}$$

Ex:- 1) The no. of defective bolts, in a box containing 'n' bolts

2) The coin tossing experiment, getting 'x' heads

3) The no. of post graduates in a group of 'n' men.

Binomial Distribution :-

It holds the following

1) Trails are repeated.

2) The no. of trails 'n' is finite and there are only '2' possible outcomes. i.e Success and failure.

3) The probability of success (p) is constant in each trial remaining constants and does not change from trial to trail.

4) The trails are Independent.

Statistical parameter of Binomial Distributions :

1) Mean :

Binomial distribution is given by

$$p(x=x) = {}^nC_x\, p^x q^{n-x} \; ; \; x = 0, 1, 2, 3, \ldots n$$

Now;

$$E(x) = \sum_{x=0}^{n} x\, p(x)$$

$$= \sum_{x=0}^{n} x\, {}^nC_x\, p^x q^{n-x}$$

$$E(x) = \sum_{x=1}^{n} x \frac{n!}{(n-x)!\, x!} p^x q^{n-x}$$

$$= \sum_{x=1}^{n} x \cdot \frac{(n-1)\, n!}{x\,(x-1)!\,\lfloor (n-1)-(x-1)} p \cdot p^{x-1} q^{(n-1)-(x-1)}$$

$$= np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!\,\lfloor (n-1)-(x-1)} p^{x-1} q^{(n-1)-(x-1)}$$

$$= np \sum_{(x-1)=0}^{n} \frac{(n-1)!}{(x-1)!\,\lfloor (n-1)(x-1)} p^{x-1} q^{(n-1)-(x-1)}$$

$$= np(p+q)^{n-1} \quad [\because p+q = 1]$$

$$= np.$$

Note :

1) $(p+q)^n = {}^nC_0\, p^0 q^n + {}^nC_1\, p^1 q^{n-1} + {}^nC_2\, p^2 q^{n-2} + \ldots$

$$\ldots + {}^nC_x\, p^x q^{n-x} + \ldots + {}^nC_n\, p^n q^0$$

$$= \sum_{x=0}^{n} {}^nC_x\, p^x q^{n-x}$$

$$E(x) = \sum_{x=0}^{n} \frac{n!}{x!\,\lfloor n-x} p^x q^{n-x}$$

2) Variance $V(x)$:

$$V(x) = E(x^2) - [E(x)]^2$$

Now, $E(x^2) = \sum_{x=0}^{n} x^2 p(x)$

$$= \sum_{x=0}^{n} [x(x-1) + x]\, p(x)$$

$$= \sum_{x=0}^{n} [x(x-1)\, p(x) + \sum_{x=0}^{n} x\, p(x)$$

$$= \sum_{x=2}^{n} x(x-1)\, {}^{n}C_x\, p^x q^{n-x} + E(x)$$

$$= \sum_{x=2}^{n} x(x-1) \frac{n!}{x!\,(n-x)!} p^x q^{n-x} + np$$

$$= \sum_{x=2}^{n} x(x-1) \frac{(n-1)n(n-2)!}{x(x-1)(x-2)!\,(n-2)-(x-2)!} p^2 q^{(n-2)-(x-2)} p^{x-2} + np$$

$$= (n-1)np^2 \sum_{n-2=0}^{n-2} \frac{(n-2)!}{(x-2)!\,(n-2)-(x-2)!} p^{x-2} q^{(n-2)-(x-2)} + np$$

$$= n(n-1)p^2 (p+q)^{n-2} + np$$

$$= n(n-1)p^2 + np \qquad [\because p+q = 1]$$

$$E(x^2) = n^2p^2 - np^2 + np$$

Now; $V(x) = E(x^2) - [E(x)]^2$

$$= n^2p^2 - np^2 + np - (np)^2$$

$$= n^2p^2 - np^2 + np - n^2p^2$$

$$= np - np^2$$

$$= np(1-p)$$

$$\boxed{V(x) = npq}$$

3) Mode :-

Mode of Binomial distribution is value of 'x' for which $p(x)$ has maximum value.

$\therefore$ Mode $=$ { integral part of $(n+1)p$; where if $(n+1)p$ is not an integral.
$(n+1)p$ and $(n+1)p-1$; if $(n+1)p$ is an integer.

4) Recurrance Relation :-

W.K.T $p(x=x) = {}^nC_x\, p^x q^{n-x}$ and $-①$

$p(x+1) = {}^nC_{x+1}\, p^{x+1} q^{n-(x+1)}$ $-②$

Now; $\frac{p(x+1)}{p(x=x)} = \frac{{}^nC_{x+1}\, p^{x+1} q^{n-(x+1)}}{{}^nC_x\, p^x q^{n-x}}$

$= \frac{\frac{n!}{(x+1)!\,(n-x-1)!}\, p^x p\, q^{n-x} q^{-1}}{\frac{n!}{x!\,(n-x)!}\, p^x . q^{n-x}}$

$\frac{p(x+1)}{p(x)} = \frac{\frac{1}{x(x+1)!\,(n-x-1)!}\,\frac{p}{q}}{\frac{1}{x!\,(n-x)!}}$

$= \frac{\frac{1}{x(x+1)!\,(n-x-1)!}\, p/q}{\frac{x!\,(n-x)(n-x-1)!}{}}$

$= \frac{1}{x!\,(x+1)(n-x-1)!}\,\frac{p}{q} \times \frac{x!\,(n-x)(n-x-1)!}{1}$

$$\boxed{\frac{p(x+1)}{p(x)} = \frac{(n-x)}{(x+1)}\,\frac{p}{q}}$$

$$\boxed{p(x+1) = \frac{(n-x)}{(x+1)}\cdot\frac{p}{q}\cdot p(x)}$$

1) 10 coins throw simanteously find the probability of getting atleast (i) 7 heads (ii) 6 heads (iii) one head.

Sol:- $P \Rightarrow$ probability for getting a head $= \frac{1}{2}$

$q = \frac{1}{2}$ $(\because p+q=1)$

Here $n = 10$.

we know that $\boxed{p(x=x) = {}^nC_x\, p^x q^{n-x}}$ $(x = 0,1,2,\ldots n)$

(i) $p(x \geq 7) = p(x=7) + p(x=8) + p(x=9) + p(x=10)$

$= p(7) + p(8) + p(9) + p(10)$

$= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{10-7} + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8}$

$+ {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^0$

$= \left(\frac{1}{2}\right)^{10} \left[{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}\right]$

$= 0.1719.$

(ii) $p(x \geq 6) = p(x=6) + p(x=7) + p(x=8) + p(x=9) + p(10)$

$= p(6) + \underbrace{p(7) + p(8) + p(9) + p(10)}_{0.1719}$

$\Rightarrow {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{10-6} + 0.1719$

$= 0.20117$

(iii) $p(x \geq 1) = 1 - p(x<1)$

$= 1 - p(x=0)$

$= 1 - {}^{10}C_6 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{10-0}$

$= 0.999$

2) If 3 of 20 tyres are defective and 4 of them are randomly chosen for inspection, what is the probability that only one of the defective tyre will be included.

$p \Rightarrow$ probability of defective tyre $= \frac{3}{20}$

$\therefore p = \frac{3}{20}$; $q = 1-p = 1-\frac{3}{20} = \frac{17}{20}$

Here ; $n = 4$

$\therefore$ probability that Exactly one tyre will be defective

$$P(X=1) = {}^4C_1 \left(\frac{3}{20}\right)^1 \left(\frac{17}{20}\right)^{4-1}$$

$$= {}^4C_1 \left(\frac{3}{20}\right)\left(\frac{17}{20}\right)^3$$

$$= 0.3685$$

3) Determine the Binomial distribution from which mean is 4 and variance is 3.

Solt Given that,

Mean $= np = 4$ —①

Varaince $= npq = 3$ —②

Since $npq = 3$

$4q = 3 \Rightarrow \boxed{q = \frac{3}{4}}$

$p = 1-q = 1-\frac{3}{4} = \frac{1}{4}$

$\boxed{p = \frac{1}{4}}$

Now, $n = ?$ subsitute The 'p' value in ①

we get

$$np = 4 \Rightarrow n\left(\frac{1}{4}\right) = 4$$

$\boxed{n = 16}$

$\therefore$ The probability $p(x=x)$

$$\therefore p(x=x) = \begin{cases} {}^{16}C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{16-x} & ; x = 0, 1, 2 \cdots n \\ 0 \; ; \text{otherwise} \end{cases}$$

4) If the probability of defect bolts is $\frac{1}{8}$ find the mean and varaince for the disturbution of defective 640

Solt $p = \frac{1}{8}$; $q = 1-p = \frac{7}{8}$; $n = 640$

$\therefore E(X) = np \Rightarrow 640 \times \frac{1}{8} = 80$.

$\therefore V(X) = npq = 640\left(\frac{1}{8}\right)\left(\frac{7}{8}\right)$

$= 70$

$\boxed{E(X) \Rightarrow \text{mean} = 80 \; ; \; V(X) = 70.}$

5) Mean and varaince of Binomial distribution if the 4 and $\frac{4}{3}$ respectively, find $p(x \geq 1)$

Solt Given that

mean $\Rightarrow np = 4$ —①

varaince $\Rightarrow npq = \frac{4}{3}$ —②

Since, $npq = \frac{4}{3}$

$4q = \frac{4}{3} \Rightarrow q = \frac{4}{3} \times \frac{1}{4} = \frac{4}{12} = \frac{1}{3}$

$\boxed{q = \frac{1}{3}}$

$p = 1-q = 1-\frac{1}{3} = \frac{2}{3} \Rightarrow \boxed{p = \frac{2}{3}}$

$npq = \frac{4}{3} \leftarrow \therefore \boxed{n = 6}$

$n\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = \frac{4}{3}$

$n\left(\frac{2}{9}\right) = \frac{4}{3}$, 4×6,

$p(x \geq 1) = 1-p(x<1)$

$= 1-p(x=0)$

$= 1-{}^nC_0\, p^0\, q^{n-0}$

$= 1-{}^6C_0\left(\frac{2}{3}\right)^0\left(\frac{1}{3}\right)^{6-0}$

$= 0.9986$

6) The mean and variance of Binomial variable (or) Distribution 'x' and parameter n, and p are 16 and 8. find $(x \geq 1)$ and $p(x \geq 2)$?

Solt Given that;

The mean and variance of a Binomial random variable 'x' are 16 and 8.

$E(x) = np = 16 - (1)$; $V(x) = npq = 8 - (2)$

Since $npq = 8 \Rightarrow 16q = 8 \Rightarrow \boxed{q = \frac{1}{2}}$

$p = 1 - q = 1 - \frac{1}{2} = \frac{1}{2} \Rightarrow \boxed{p = \frac{1}{2}}$

$np = 16 \Rightarrow n\left(\frac{1}{2}\right) = 16$

$\boxed{n = 32}$

(i) $p(x \geq 1) \Rightarrow 1 - (x < 1) \Rightarrow 1 - p(x = 0)$

$= 1 - {}^{n}C_0\, p^0\, q^{n-0}$ $\quad 1 - {}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32} \Rightarrow 1 - 1\left(\frac{1}{2}\right)^{32}$

$= 1 - {}^{32}C_0\, p^0\, q^{32-0} = 0.999$

(ii) $P(X \geq 2) \Rightarrow 1 - p(x < 2)$

$1 - p(x=0) + p(x=1)$

$1 - {}^{32}C_0\, p^0 q^{32-0} + {}^{32}C_1\, p^1 q^{32-1}$

$= 1 - 0.999 + 32\left(\frac{1}{2}\right)^{16}$

$= 1 - 0.999 + 16 = 16.001$

7) The mean and variance of Binomial distribution are 2 and $\frac{2}{5}$ and then find n.

Solt Given that $E(x) = 2$; $V(x) = \frac{2}{5}$; $n = ?$

$np = 2 - (1)$; $npq = \frac{2}{5} - (2)$

Since $npq = \frac{2}{5} \Rightarrow 2(q) = \frac{2}{5}$

$q = \frac{2}{5 \times 2} = \frac{2}{10} = \frac{1}{5}$

$\boxed{q = \frac{1}{5}}$

$p = 1 - q = 1 - \frac{1}{5} = \frac{4}{5}$

Now; $np = 2 \Rightarrow n\left(\frac{4}{5}\right) = 2$

$n = 2 \times \left(\frac{5}{4}\right) = \frac{5}{2}$

$\boxed{n = 2.5}$

8) Two dice are thrown 5 times find the probability of getting 7 as sum. i) at least once

ii) exactly two times

iii) $P(1 < x < 5)$

Solt Let 'p' be the probability of getting sum 7 if two dice are thrown, favorable cases are

(1,6) (6,1) (2,5) (5,2) (3,4) (4,3)

$\Rightarrow$ 6 favorable cases out of 36. i.e $\frac{6}{36}$

$p = \frac{6}{36} = \frac{1}{6}$

$q = 1 - p = 1 - \frac{1}{6} = \frac{5}{6}$

Here $n = 5$

(i) atleast once.

$p(x \geq 1) = 1 - p(x < 1)$

$= 1 - p(x = 0)$

$= 1 - {}^{5}C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^{5-0}$

$= 0.5981$

ii) $p(x = 2) = {}^{5}C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^{5-2}$

$5 \times 10 \left(\frac{1}{36}\right)\left(\frac{5}{6}\right)^3 \Rightarrow \frac{5}{3}\left(\frac{5}{6}\right)^3$

$= 0.1607$

iii) $p(1 < x < 5)$

$p(x=2) + p(x=3) + p(x=4)$

$= {}^{5}C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^{5-2} + {}^{5}C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{5-3} + {}^{5}C_4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^{5-4}$

$= 0.1961$

9) Out of 800 families, of 5 children each, How many a) 3 boys
b) 5 girls
c) either 2 or 3 boys.
d) atleast 3 boys.

Sol:- Assume that, equal probabilities for boys and girls i.e $p = \frac{1}{2}$, $q = \frac{1}{2}$; $n = 5$; X = no. of boys in each family.

(i) $P(X=3)$ (3 boys)

$= {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3} \Rightarrow \frac{5}{16} = 0.3125$

$10\left(\frac{1}{8}\right)\left(\frac{1}{2}\right)^2 \Rightarrow \frac{10}{8}\left(\frac{1}{4}\right) = \frac{5}{16} =$

No. of families having 3 boys $= 800 \times \frac{5}{16}$

$= 250.$

∴ out of families, 250 families having 3 boys.

ii) 5 girls

P(no boys) = $p(X=0) \Rightarrow p$(5 girls)

$\therefore p(X=0) = {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0}$

$= {}^5C_0 \left(\frac{1}{2}\right)^5$

=

Out of 800 families 25 families having 5 girls.

iii) p(either 2 or 3 boys) = 1

$p(X=2) + p(X=3)$

${}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2} + {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3}$

$= \frac{15}{8} + \frac{5}{16} =$

Out of 800, families either 2 or 3 boys

800 x

d) atleast 3 boys.

$p(X \geq 3) = 1 - p(X<3)$

$= 1 - p(X=3) \Rightarrow 1 - [p(0) + p(1) + p(2)]$

$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0} + {}^5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{5-1}$

$+ {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2}$

$= 775$

Out of families, 775 families atleast 3 boys.

10) Assume that 50 % of Engineering students are good in mathematics determine the prob. that among 180 Engineering students.

i) Exactly 10 $p(x=10)$
ii) Atleast 10 $p(x \geq 10)$
iii) atmost 8 $p(x \leq 8)$
iv) atleast 2 and atmost 9 are good in Mathematics. $p(2 \leq x \leq 9)$

Sol:- Let 'x' be the no. of students good in mathemat

p = probability of students good in mathematis

$p = 50\% = \frac{1}{2}$; $q = 1 - p = \frac{1}{2}$; $n = 180$

i) Exactly 10 students : $p(x=10)$

${}^{180}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{180-10} \Rightarrow {}^{180}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{170}$

$\simeq 0.1669$

ii) Atleast 10 $\Rightarrow p(x \geq 10)$.

$p(X \geq 10) = 1 - p(X \leq 9)$

$1 - p(9) + p(8) + p(7) + p(6) + p(5) + p(4) + p(3) + p(2)$

$+ p(1)$

$= 1 - \sum_{x=0}^{9} {}^{180}C_x \left(\frac{1}{2}\right)^{180}$

$= \left(\frac{1}{2}\right)^{18} (18C_0 + \cdots {}^{18}C_{18})$

~~$P(X \geq 10) \simeq 0.9999$~~

iii) Atmost '8' students $\Rightarrow P(X \leq 8)$

$P(0) + P(1) + P(2) + P(3) + \cdots + P(8)$

$$\Rightarrow \sum_{x=0}^{8} {}^{18}C_x \left(\frac{1}{2}\right)^{18}$$

$$= \left(\frac{1}{2}\right)^{18} \left[{}^{18}C_0 + {}^{18}C_1 + {}^{18}C_2 + \cdots + {}^{18}C_8\right]$$

iv) $P(8 \leq x \leq 9) \Rightarrow \sum_{x=2}^{9} {}^{18}C_x \left(\frac{1}{2}\right)^{18}$

$$\left(\frac{1}{2}\right)^{18} \left[{}^{18}C_2 + {}^{18}C_3 + \cdots + {}^{18}C_9\right]$$

iv) $P(2 \leq x \leq 9) \Rightarrow P(2) + P(3) + P(4) + P(5) + P(6) + P(7) + P(8) + P(9)$

$$= {}^{80}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{80-2} + {}^{80}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{80-3} + {}^{80}C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{80-4} + {}^{80}C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{80-5} + {}^{80}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{80-6} + {}^{80}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{80-7} + {}^{80}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{80-8} + {}^{80}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{80-9}$$

$$= 2.1870 \times 10^{-13}$$

*** Imp.

11) fit a binomial distribution to the following frequency distribution.

x	0	1	2	3	4	5	6
f(x)	13	25	52	58	32	16	4

Sol^r Here $n = 6$ (no. of trails)

$$N = \sum f_i = 13 + 25 + 52 + 58 + 32 + 16 + 4$$

$$= 200$$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{(0 \times 13) + 1(25) + 2(52) + \cdots + 6(4)}{200}$$

$$= 2.675$$

Now, Mean of the Binomial distribution $= np$.

$$np = 2.675$$

$$6p = 2.675$$

$$p = 0.446$$

$$q = 0.554$$

Hence, the binomial distribution is fitted given by

$$N(q+p)^n$$

$$N(q+p)^n = 200[0.55 + 0.446]^6$$

$$= 200\left[{}^6C_0 (0.554)^6 + {}^6C_1 (0.554)^5 (0.446) + {}^6C_2 (0.554)^4 (0.446)^2 + {}^6C_3 (0.554)^3 (0.446)^3 + {}^6C_4 (0.554)^2 (0.446)^4 + {}^6C_5 (0.554)(0.446)^5 + {}^6C_6 (0.446)^6\right]$$

$$= 5.788 + 27.92 + 56.18 + 60.32 + 36.42 + 11.728 + 1.5732.$$

The repeated frequencies can be rounded off to the nearest integers to get expected frequencies as whole numbers.

The successful terms in the Expansion gives The Expected or theoritical frequencies which are

x	0	1	2	3	4	5	6
f(x)	13	25	52	58	32	16	4
Expected frequencies	6	28	56	60	36	12	2

12) Binomial distribution to the following frequency distribution.

x	0	1	2	3	4	5
f(x)	2	14	20	34	22	8

Sol: $n=5$ there $N = \sum f_i = 2+14+20+34+22+8 = 100$.

$\bar{x} \neq \sum x f(x)$

Mean $\Rightarrow \dfrac{\sum f_i x_i}{\sum f_i} = \dfrac{(0)(2) + 1(14) + 2(20) + 3(34) + 4(22) + 5(8)}{100}$.

Mean $\Rightarrow \dfrac{284}{100} = 2.84$

Now, mean of the binomial distribution $= np$.

$n=5 \Rightarrow np = 2.84$

$5p = 2.84 \Rightarrow p = 2.84/5$

$p = 0.568$

$q = 1-p \Rightarrow 1-0.568$

$q = 0.432$

Hence, the binomial distribution is fitted given by

$N(q+p)^n$

$N(0.568 + 0.432)^5 \Rightarrow 100(0.568+0.432)^5$

$100[{}^5C_0 (0.568)^5 + {}^5C_1 (0.568)^4 (0.432)^1 + {}^5C_2 (0.568)^3 (0.432)^2 + {}^5C_3 (0.568)^2 (0.432)^3 + {}^5C_4 (0.568)^1 + {}^5C_5 (0.568)^0 (0.432)^5$.

$\Rightarrow 1.505 + 9.891 + 26.010 + 34.199 + 22.483 + 5.912$. These are the theoritical frequencies

x	0	1	2	3	4	5
f(x)	2	14	20	34	22	8
Expected values	2	10	26	34	22	6

Poisson Distribution :

(Binomial appox. to poisson Distribution).

The poisson Distribution can be derived as a limiting case of the binomial distribution under the following conditions.

1) p is very small i.e $p \to 0$

2) n is very very large. i.e $n \to \infty$

where 'n' is the no. of trials.

3) $np = \lambda$ (say), λ is called the parameter of the poisson distribution.

Derivation :

W.K.T $P(X=x) = {}^nC_x\, p^x q^{n-x}$

(or)

$$P(X=r) = {}^nC_r\, p^r q^{n-r}$$

$$= \frac{n!}{r!\,(n-r)!}\, p^r q^{n-r}$$

$$\frac{n(n-1)(n-2)\cdots(n-r-1)(n-r)!}{r!\,(n-r)!}\, p^r (1-p)^{n-r}$$

Since $np = \lambda$

$$\Rightarrow p = \frac{\lambda}{n} \quad \& \quad n = \frac{\lambda}{p}$$

$$P(X=r) = \frac{\frac{\lambda}{p}\left(\frac{\lambda}{p}-1\right)\left(\frac{\lambda}{p}-2\right)\cdots\left(\frac{\lambda}{p}-r-1\right) p^r \frac{(1-p)^n}{(1-p)^r}}{r!}$$

$$= \frac{\frac{\lambda}{p}\left(\frac{\lambda-p}{p}\right)\left(\frac{\lambda-2p}{p}\right)\cdots\left(\frac{\lambda-p(r-1)}{p}\right) p^r \cdot \frac{(1-p)^n}{(1-p)^r}}{r!}$$

$$P(X=r) = \frac{\lambda(\lambda-p)(\lambda-2p)\cdots(\lambda-p(r-1))}{p^r \cdot r!}\, p^r \frac{(1-p)^n}{(1-p)^r}$$

$$\lim_{n\to\infty} P(r) = \lim_{n\to\infty} \frac{\lambda(\lambda-p)(\lambda-2p)\cdots(\lambda-p(r-1))}{r!}$$

$$P(r) = \frac{\lambda(\lambda-0)(\lambda-0)\cdots(\lambda-0)}{r!}\, \frac{e^{-\lambda}}{(1-0)^r}$$

$$\boxed{P(X=r) = \frac{\lambda^r}{r!} e^{-\lambda}}$$

$\lim_{n\to\infty} \frac{(1-\lambda/n)^n}{(1-\lambda/n)^r}$

$\left(\because \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e^1\right)$

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}\ ; \ x = 0,1,2,3\cdots n$$

$$\boxed{P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}}$$

$$P(X=x) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!}\ ; & x = 0,1,2\cdots n \\ 0\ ; & \text{otherwise} \end{cases}$$

def of poission dist^n :

A R.V 'X' is said to follow a poission dist^n if the assumes only non-negitive values and its probability density fun^n is given by

$$P(X=x) = P(x,\lambda) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!}\ ; & x = 0,1,2\cdots n. \\ 0\ ; & \text{otherwise} \end{cases}$$

Here $\lambda > 0$ is the parameter of the distribution.

Exp 1) The no. of defective bulbs manufactured by reputed company.

2) The no. of telephone calls at a minute and a swith board.

3) The no. of cars crossing a certain point in one minute.

Constance of poisson Distribution :

1) Mean :

$$E(x) = \sum_{x=0}^{\infty} x\, p(x)$$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\lambda}.\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{x\,\lambda^x}{x!} \Rightarrow e^{-\lambda} \sum_{x=1}^{\infty} \frac{x\,\lambda^x}{x(x-1)!}$$

$$= e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda.\lambda^{x-1}}{(x-1)!} \Rightarrow \lambda.e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$E(x) = \boxed{\lambda e^{-\lambda} e^{\lambda}}$$

$$\boxed{E(x) = \lambda} \Rightarrow \text{where } \lambda = np.$$

2) Variance :

$$V(x) = E(x^2) - [E(x)]^2$$

$$= \sum_{x=0}^{\infty} x^2 p(x) - \lambda^2$$

$$= \sum_{x=0}^{\infty} [x(x-1) + x]\, p(x) - \lambda^2$$

$$= \sum_{x=0}^{\infty} x(x-1)\, p(x) + \sum_{x=0}^{\infty} x\, p(x) - \lambda^2$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda}\lambda^x}{x!} + E(x) - \lambda^2$$

$$= e^{-\lambda} \sum_{x=2}^{\infty} \frac{x(x-1)\,\lambda^x}{x(x-1)(x-2)!} + \lambda - \lambda^2$$

$$= e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^2 \lambda^{x-2}}{(x-2)!} + \lambda - \lambda^2$$

$$= e^{-\lambda} \lambda^2 e^{\lambda} + \lambda - \lambda^2$$

$$= \lambda^2 + \lambda - \lambda^2 = \lambda$$

$$\boxed{V(x) = \lambda} \rightarrow \lambda = np$$

3) Mode :

Mode is the value of 'x' for which the probability p(x) is maximum.

$\therefore p(x) \geq p(x-1)$

$p(x) \geq p(x+1)$

Now; $p(x) \geq p(x+1)$

$$\frac{e^{-\lambda}\lambda^x}{x!} \geq \frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!}$$

$$\frac{1}{x!} \geq \frac{\lambda}{(x+1)!}$$

$$\frac{1}{x!} \geq \frac{\lambda}{(x+1)x!}$$

$$\Rightarrow x+1 \geq \lambda$$

$$x \geq \lambda - 1 \quad \text{---(1)}$$

and $p(x) \geq p(x-1)$

$$\therefore x \leq \lambda \quad \text{---(2)}$$

From (1) and (2), we have $\lambda - 1 \leq x \leq \lambda$.

~~case 1)~~ The mode of the p.D lies b/n λ and $\lambda - 1$

case 1) If 'λ' is an integer then $\lambda - 1$ is also integer. So, we have 2 max. values and the distribution is Binomial and the two modes are $\lambda - 1$ and λ.

case 2 : If the 'λ' is not an integer, the mode of the poisson distribution is integral part of 'λ'.

4) Recurrence Relation :-

we have $p(x) = \frac{e^{-\lambda}\lambda^x}{x!}$ —①

$$p(x+1) = \frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!} \quad —②$$

② ÷ ①

$$\frac{p(x+1)}{p(x)} = \frac{\frac{e^{-\lambda}\cdot\lambda^{x+1}}{(x+1)!}}{\frac{e^{-\lambda}\cdot\lambda^{x}}{x!}}$$

$$\frac{p(x+1)}{p(x)} = \frac{\lambda}{(x+1)}$$

$$\boxed{p(x+1) = \frac{\lambda}{x+1}\,p(x)} \quad (or)$$

$$\boxed{p(x) = \frac{\lambda}{x}\,p(x-1)}$$

Problems :-

1) A Hosptial switch board recieves on avg. of 4 Emergency calls in a 10 minute interval. what is probability that

i) at most two Emergency calls in 10 minute intervals. (There are)

ii) There are Extactly 3 Emergency calls in 10 min. intervals.

Soln Given that, λ = 4 calls from 10 minutes

$$\lambda = 4$$

and $p(x=x) = \frac{e^{-\lambda}\cdot\lambda^x}{x!}$

P(at most 2 calls) = $p(X \le 2)$

$= p(x=0) + p(x=1) + p(x=2)$

$= \frac{e^{-\lambda}\lambda^0}{0!} + \frac{e^{-\lambda}\lambda^1}{1!} + \frac{e^{-\lambda}\lambda^2}{2!}$

$= e^{-4} + \frac{e^{-4}\cdot 4}{1!} + \frac{e^{-4}\cdot 4^2}{2!}$

$= e^{-4}\left[1 + 4 + \frac{4^2}{2!}\right]$

$= 0.2381$ //

ii) probability of Extactly '3' Emergenar calls.

$\because \lambda = 4$

$p(x=x) = \frac{e^{-\lambda}\cdot\lambda^x}{x!}$

$x=3 \Rightarrow p(x=3) = \frac{e^{-4}(4^3)}{3!}$

$= \frac{e^{-4}\times 64}{6} = 0.1954$

$p(x=3) \simeq 0.1954$

2) It has been found that 2% of the tools produced by certain machine are defective. what is the probability that in a shipment of 400 such that.

a) 3% or more b) 2% or less will prove defective.

Soln Given that n=400.

let p = probability of defective tool

$p = 2\% = 0.02$

$\therefore \lambda = np \Rightarrow 400(0.02) = 8$ //

(mean no. of defective tools)

w.k.T $p(x=x) = \frac{e^{-\lambda}\cdot\lambda^x}{x!}$

i) $P(X \geq 3\%) = P\left(X \geq \frac{3}{100} \times 400\right)$

$= P(X \geq 12)$

$= 1 - P(X < 12)$

$= 1 - [P(X=0) + P(X=1) + \cdots + P(X=11)]$

$= 1 - \left(e^{-8}\frac{8^0}{0!} + e^{-8}\frac{8^1}{1!} + e^{-8}\frac{8^2}{2!} + \cdots + \frac{e^{-8}\cdot 8^{11}}{11!}\right)$

$= 0.1119$

ii) $n = 400 ; p = 2$

$\lambda = np \Rightarrow 400 \times (2\text{on}) = 8$

$\Rightarrow 2\%$ of $400 = 8$

$P(X \leq 8) \Rightarrow P(x \leq 8) = \sum_{x=0}^{8} \frac{e^{-8}\, 8^x}{x!}$

$P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6) + P(7)$

$\Rightarrow e^{-0}\frac{0^0}{0!} + e^{-1}\frac{1}{1!} + e^{-2}\frac{2}{2!} + e^{-3}\frac{3}{3!} + \cdots e^{-7}\frac{7}{7!}$

$P(x \leq 8) \simeq 0.5925$

3) If a R.V as poission distribution such that $P(1) = P(2)$ Find (i) mean of the distribution.

(ii) $P(4)$

(iii) $P(X \geq 1)$

(iv) $P(1 < x < 4)$

Solr Given that $P(1) = P(2)$

i) $\frac{e^{-\lambda}\cdot\lambda^1}{1!} = \frac{e^{-\lambda}\cdot\lambda^2}{2!}$

$\Rightarrow 1 = \frac{\lambda}{2} \Rightarrow \boxed{\lambda = 2}$

Mean of the distribution is 2

ii) $P(4) \Rightarrow P(x=4)$

$\frac{e^{-\lambda}\cdot\lambda^4}{4!}$

$= \frac{e^{-2}\, 2^4}{4!} \Rightarrow 0.09022$

iii) $P(X \geq 1) = 1 - P(X < 1)$

$\Rightarrow 1 - P(X=0)$

$= 1 - \frac{e^{-\lambda}\cdot\lambda^0}{0!} = 1 - e^{-2}$

$= 0.8647$

iv) $P(1 < x < 4)$

$P(x=2) + P(x=3)$

$\Rightarrow e^{-2}\frac{2^2}{2!} + e^{-2}\frac{2^3}{3!} \Rightarrow 2e^{-2} + \frac{4}{3}e^{-2}$

$= \frac{10}{3}e^{-2} \Rightarrow \simeq 0.4511$

4) 2% of the items of a factory are defective. the items are packed in boxes what is the prob. there will be

i) Two defective items

ii) Atleast 3 defective items in a box of 100 items

Solr Given that

$n = 100$; p = prob. of defective items.

$p = 2\% = 0.02$

$\Rightarrow \lambda = np = 100 \times 0.02$

$\boxed{\lambda = 2}$

(i) $P(x=2)$

$\frac{e^{-\lambda}\lambda^2}{2!} \Rightarrow \frac{e^{-2}\cdot 2^2}{2!} = 0.2706$

ii) $P(X \geq 3) = 1 - P(X < 3)$

~~$1 - e^{-\lambda}\lambda^2 + 1$~~

~~$3!$~~

$1 - p(x=0) + p(x=1) + p(x=2) + 4$

$\Rightarrow 1 - e^{-2} + 2e^{-2} + 2e^{-2}$

$1 - 5e^{-2} \simeq 1 - 5(0.1353)$

$= 0.3233$

5) Using Recurrance relation formula find the probabilities when $x = 0, 1, 2, 3, 4$ and 5, if the mean of the p.D is 3.

Solr Recurrance relation formula $\boxed{p(x+1) = \frac{\lambda}{x+1} p(x)}$

~~$p(1) = \frac{\lambda}{1} p(0)$~~

~~$p(2) = \frac{\lambda}{2} p(1)$~~

Given that mean of the p.D is 3

i.e $\lambda = 3$.

W.K.T $\Rightarrow p(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

$\therefore p(X=0) = \frac{e^{-3} \cdot 3^0}{0!} \Rightarrow e^{-3}$

$= 0.0498$.

By reference relation formula, we have

$\boxed{p(x+1) = \frac{\lambda}{x+1} p(x)}$

$p(x+1) = \frac{3}{x+1} p(x)$ — ①

put $x=0$ in eq ①, we get

$p(1) = \frac{3}{1} p(0)$

$= 3(0.0498)$

$= 0.1494$.

put $x=1$ in eq ① we get

$p(2) = \frac{3}{2}(0.1494)$

$= 0.2241$

put $x=2 \Rightarrow p(3) = \frac{3}{3} p(2)$

$= 0.2241$

put $x=3 \Rightarrow p(4) = \frac{3}{4} p(3)$

$= \frac{3}{4}(0.2241)$

$= 0.1681$

put $x=4 \Rightarrow p(5) = \frac{3}{5} p(4)$

$\Rightarrow \frac{3}{4}(0.1681) = 0.1009$

put $x=5 \Rightarrow p(6) \Rightarrow \frac{3}{6} p(5)$

$\Rightarrow \frac{3}{4}(0.1009) = 0.075675$

6) If a passon Distribution is such that $p(x=1)\frac{3}{2} = p(x=3)$ Find i) $p(x \geq 1)$ ii) $p(x \leq 3)$ iii) $p(2 \leq x \leq 5)$

Solr Given that $p(x=1)\frac{3}{2} = p(x=3)$

$\frac{3}{2} \frac{e^{-\lambda} \cdot \lambda^1}{1!} = \frac{e^{-\lambda} \cdot \lambda^3}{3!}$

$\Rightarrow \frac{3}{2} = \frac{\lambda^2}{6} \Rightarrow \lambda^2 = 9$

$\Rightarrow \boxed{\lambda = 3}$

(i) $p(x \geq 1) = 1 - p(x<1)$

$= 1 - p(x=0)$

$= 1 - \frac{e^{-\lambda} \cdot \lambda^0}{0!} \Rightarrow 1 - e^{-3}$

$= 0.950213$

ii) $p(x \leq 3) \Rightarrow$ ~~$p(x=3)$~~

$\Rightarrow p(0) + p(1) + p(2) + p(3)$

$P(x \leq 3) \Rightarrow 0.0498 + 0.1494 + 0.2241 + 0.2241$

$= 0.6474$

iii) $P(2 \le x \le 5) = p(2) + p(3) + p(4) + p(5)$

$\Rightarrow$ W.K.T $p(x=2)$ and $p(x=3)$

$$p(x=4) \Rightarrow \frac{e^{-3}.3^4}{4!} = \frac{81}{24} e^{-3} = 3.375\, e^{-3}$$

$$p(x=5) \Rightarrow \frac{e^{-3}.3^5}{5!} = \frac{243}{120} e^{-3} = 2.025\, e^{-3}$$

$\Rightarrow (4.5 + 4.5 + 3.375 + 2.025)e^{-3} = 14.4\, e^{-3}$.

$P(2 \le x \le 5) \simeq 0.7171$

7) ~~Quick~~ a p.D following data and calculate the expected frequencies.

x	0	1	2	3	4
$f(x)$	109	65	22	3	1

Soln Here $N = \sum f_i = 109 + 65 + 22 + 3 + 1$

$= 200$.

Mean $\Rightarrow \frac{\sum x_i f_i}{\sum f_i} = \frac{(0 \times 109) + (1 \times 65) + (2 \times 22) + (3 \times 3) + (4 \times 1)}{200}$.

$\lambda = 0.61$

Mean of p.D $\lambda = 0.61$

Hence, The theoritical frequencies are given by

$N.p(x)$ where $x = 0, 1, 2, 3, 4$.

$$x=0 \Rightarrow Np(0) = 200 \frac{e^{-\lambda}.\lambda^0}{0!} = 200\, e^{-0.61}$$

$$= 108.67 \simeq 109$$

$$x=1 \Rightarrow Np(1) = 200 \cdot \frac{e^{-\lambda}.\lambda^1}{1!} = 200\, e^{-0.61}(0.61)$$

$$= 66.29 \Rightarrow 66$$

$$x=2 \Rightarrow Np(2) = 200 \frac{e^{-\lambda}.\lambda^2}{2!} = 20.22 = 20.$$

$$x=3 \Rightarrow Np(3) = 200 \cdot \frac{e^{-\lambda}\lambda^3}{3!} = 4.11 \simeq 4$$

$$x=4 \Rightarrow Np(4) = 200 \cdot \frac{e^{-\lambda}\lambda^4}{4!} = 0.63 \simeq 1$$

observed freq.	109	65	22	3	1
expected freq.	109	66	20	4	1

8) If a p.D following data.

x	0	1	2	3	4	5
$f(x)$	142	156	69	27	5	1

Soln $\sum f_i = N = 142 + 156 + 69 + 27 + 5 + 1 = 400$.

$$\lambda = \frac{\sum x_i f_i}{\sum f_i} = \frac{0(142) + 1(156) + 2(69) + 3(27) + 4(5) + 5(1)}{400}$$

$$= \lambda = \frac{400}{400} = 1$$

$$\boxed{\lambda = 1}$$

Hence, the theoritical frequencies are given by

$N.p(x)$ where $x = 0, 1, 2, 3, 4, 5$.

$$x=0 \Rightarrow Np(0) \Rightarrow 400 \frac{e^{-\lambda}.\lambda^0}{0!} = 400\, e^{-1}$$

$$\Rightarrow 147.15 \simeq 147$$

$$x=1 \Rightarrow Np(1) \Rightarrow 400 \frac{e^{-\lambda}.\lambda^1}{1!} = 400\, e^{-1}(1)$$

$$= 147.15 \simeq 147$$

$$x=2 \Rightarrow Np(2) \Rightarrow 400\, e^{-1} \frac{1^2}{2!} \Rightarrow 73.58$$

$$\Rightarrow 73.58 \simeq 74$$

$x=3 \Rightarrow Np(3) = 400\,e^{-1}\frac{1^3}{3!} \Rightarrow 24.53 \simeq 25$

$x=4 \Rightarrow Np(4) = 400\,e^{-1}\frac{1^4}{4!} \Rightarrow 6.13 \simeq 6$

$x=5 \Rightarrow Np(5) = 400\,e^{-1}\frac{1^5}{5!} \Rightarrow 1.23 \simeq 1$

observed freq.	142	156	69	27	5	1
expected freq.	147	147	74	25	6	1

(1) Gamma fun^n : $\int_0^\infty e^{-x}x^{n-1}\,dx = \sqrt{n}$

* $\sqrt{n+1} = n!$
* $\sqrt{n+1} = n\sqrt{n}$ (or) $\sqrt{n} = (n-1)\sqrt{n-1}$
* $\sqrt{1/2} = \sqrt{\pi}$

Normal dist^n : mean = median = mode

(2) Mean, Variance, Mode, median properties of N.D

3. Continous probability distribution.

- Normal Distribution :

Cheif characteristics of Normal dist^n (or) properties of Normal dist^n.

- The curve is bell shaped and symmetrical about the line $X = \mu$.

Area under the normal curve :

The prob. that R.V 'x' will lie b/n $x=\mu$ and $x=x_1$ is given by $p(\mu < x < x_1) = \int_\mu^{x_1} f(x)\,dx = \frac{1}{\sigma\sqrt{2\pi}}\int_\mu^{x_1} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$

★ How to find prob. density of Normal curve.

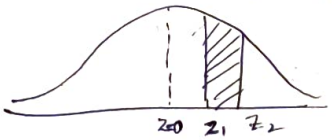
The prob. that the normal variate 'x' with mean μ and standard deviation σ lies b/n two specific values x_1 and x_2 with $x_1 \le x_2$ can be obtained using area under the standard normal curve as follows :

Step 1 : perform the change of scale $z = \frac{x-\mu}{\sigma}$ and find z_1 and z_2 corresponding values of x_1 & x_2 resp^y

Step 2 : TO find $P(x_1 \le x \le x_2) = P(z_1 \le z \le z_2)$

case 1 : If both z_1 and z_2 are positive (or both negitive), then $P(x_1 \le X \le x_2) = |A(z_2) - A(z_1)|$

= Area under the normal curve from 0 to z_2) − Area under the nrml curve from 0 to z_1)



$P(0 < z < z_2) - P(0 < z < z_1)$

(or)

$= |A(z_2) - A(z_1)|$

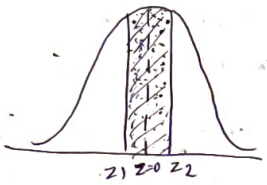
and $P(x_1 < x < x_2) = P(z_1 < z < z_2)$



$= P(0 < z < -z_2) - P(0 < z < -z_1)$

$= P(0 < z < z_2) - P(0 < z < z_1)$

case 2:- If $z_1 < 0$ and $z_2 > 0$ then $P(x_1 \le x \le x_2) = A(z_2) + A(z_1)$



$= P(z_1 < z < 0) + P(0 < z < z_2)$

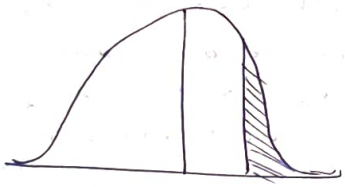
(or)

$A(z_1) + A(z_2)$

Step 2:- To find $P(z > z_1)$

case 1:- If $z_1 > 0$ then

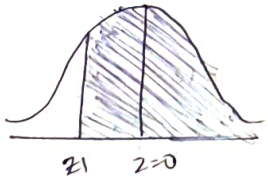
$P(z > z_1) = 0.5 - A(z_1)$ [$\because P(z<0) = P(z>0) = \frac{1}{2}$]



$P(z > z_1) = 0.5 - P(0 < z < z_1)$

$= 0.5 - A(z_1)$

case 2:- If $z_1 < 0$

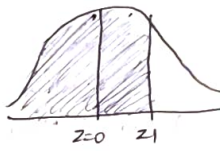


$P(z > z_1) = 0.5 + P(-z_1 < z < 0)$

$= 0.5 + P(0 < z < z_1)$

Step 3:- To find $P(z < z_1)$

case 1:- If $z_1 > 0$



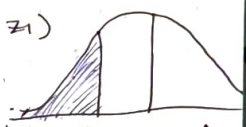
$P(z < z_1) = 0.5 + P(0 < z < z_1)$

$0.5 + A(z_1)$

case 2:- If $z_1 < 0$

$P(z < z_1) = 0.5 - P(-z_1 < z < 0)$

$= 0.5 - P(0 < z < z_1)$



1) For a Normal dist^n variance (variante) with mean 1 and S.D 3. find the prob. that

i) $3.43 \le x \le 6.19$

ii) $-1.43 \le x \le 6.19$

Sol:- Given $\mu = 1$, $\sigma = 3$

let $z = \frac{x-\mu}{\sigma}$

(i) when $x = 3.43 \Rightarrow z = \frac{3.43 - 1}{3} = 0.81 = z_1$ (say)

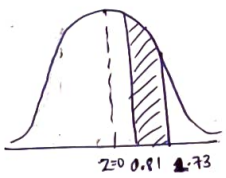
when $x = 6.19 \Rightarrow z = \frac{6.19 - 1}{3} = 1.73 = z_2$ (say)

$P(3.43 \le x \le 6.19) = P(0.81 \le z \le 1.73)$

$= P(0 < z < 1.73) - P(0 < z < 0.81)$

$= 0.4582 - 0.2910$

$= 0.1672$



ii) when $x = -1.43 \Rightarrow z = \frac{-1.43 - 1}{3}$

$= -0.81$

when $x = 6.19 \Rightarrow z = \frac{6.19 - 1}{3} = 1.73$

$\therefore P(-1.43 \leq x \leq 6.19) = P(-0.81 \leq z \leq 1.73)$

$= P(-0.81 < z < 0) + P(0 < z < 1.73)$

$P(0 < z < +0.81) + P(0 < z < 1.73)$

$0.2910 + 0.4582 =$



2) For a Normal dist^n, variance = mean = 30, SD = 5 find

i) $P(26 \leq x \leq 40)$

ii) $P(x \geq 45)$

Sol^n G.T $\mu = 30$; $\sigma = 5$

let $z = \frac{x-\mu}{\sigma}$

(i) when $x = 26 \Rightarrow \frac{26-30}{5} = -0.8 = z_1$ (say)

when $x = 40 \Rightarrow \frac{40-30}{5} = 2 = z_1$ (say)

$\Rightarrow P(-0.8 \leq z \leq 2)$

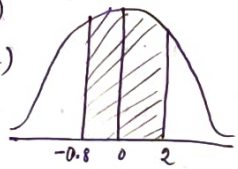
$\therefore P(26 \leq x \leq 40) = P(-0.8 < z < 2)$

$= P(-0.8 < z < 0) + P(0 < z < 2)$

$= P(0 < z < +0.8) + P(0 < z < 2)$

$= 0.4772 + 0.2881$

$= 0.7653$



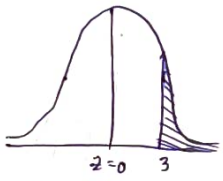
ii) when $x = 45 \rightarrow z = \frac{45-30}{5} = 3$

$P(x \geq 45) = P(z \geq 3)$

$0.5 - P(0 < z < 3)$

$= 0.5 - 0.49865$

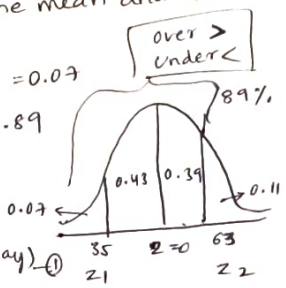
$= 0.000135$



3) In a Normal dist^n, 7% of items are under 35 % and 89 % are under 63. Determine the mean and variance of the distribution.

Sol^n Given that $P(X < 35) = 7\% = 0.07$

$P(x < 63) = 89\% = 0.89$



from the figure

$x = 35 \Rightarrow z = \frac{x-\mu}{\sigma} \Rightarrow \frac{35-\mu}{\sigma} = z_1$ (say) — (1)

when $x = 63 \Rightarrow z = \frac{63-\mu}{\sigma} = z_2$ (say) — (2)

from the figure, we have

$P(-z_1 < z < 0) = 0.43$

$P(0 < z < z_1) = 0.43$

$z_1 = 1.48$

and $P(0 < z < z_2) = 0.39$

$z_2 = 1.23$

$\Rightarrow$ Sub The z_1 and z_2 value in eq^n (1) & (2)

we get

$\frac{35-\mu}{\sigma} = -1.48$

$= -1.48\sigma + \mu = 35$ — (A)

and $\frac{63-\mu}{\sigma} = 1.23$

$\Rightarrow 1.23\sigma + \mu = 63$ — (B)

Solving (A) and (B) we get

$\mu = 50.3$ and $\sigma = 10.332$